A Model of Monetary Singleness *

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Abstract

New forms of private digital money raise questions about monetary singleness, whereby different forms of money are valued at par and can be used regardless of issuer. This paper develops a three-period banking model in which issuers choose their liabilities' repayment baskets and transferability. The model shows that par valuation is neither necessary nor sufficient for efficient investment. Even within a common nominal currency, differences between repayment promises and issuer resources create settlement burdens that constrain investment as they increase, leading to transfer fees and eventually preventing payments using another issuer's liabilities. Coordination on a shared unit of account extends exchange without fees but can leave investment constrained when the shared repayment basket is poorly aligned with issuer resources. Central bank reserves can support efficient exchange without fees, while cash provides an alternative public settlement channel that preserves opportunities for trade.

Keywords: Banking, Money, Singleness, Unit of account

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1. Introduction

A central feature of modern monetary systems is that different forms of money are expected to be interchangeable at par. Commercial bank deposits generally meet this expectation within a framework of bank regulation and interbank settlement. This framework has evolved over time, and the growth of privately issued digital money is bringing renewed attention to the conditions needed to sustain par valuation and interoperability. For example, two stablecoins may both be denominated in US dollars and intended to exchange at par. Users would therefore expect them to have equal value and to be usable on the same terms. The two liabilities may nevertheless not exchange one-for-one, and differences across issuers can make payments more costly or restrict where their liabilities circulate. Monetary singleness describes a system in which different forms of money exchange at par and circulate without regard to their issuer. This paper asks when differences across issuers erode monetary singleness and whether the resulting departures from par reduce efficiency. The analysis shows that parity across forms of private money is neither necessary nor sufficient for efficient trade. Limited departures from par can be accommodated, while sufficiently large differences in issuer resources and repayment promises constrain investment and can undermine interoperability. When these differences impose substantial settlement costs, policy intervention can reduce the resulting fragmentation.

Even when liabilities are valued at par, transferring them may involve fees or settlement restrictions. Fees may be known in advance, while the cost or availability of settlement can depend on conditions at the time of payment. Departures from par can also reflect uncertainty about the quality of backing assets or limits to redemption. Ahmed, Aldasoro, and Duley (2024) examine stablecoin depegging episodes associated with concerns about reserve assets. Bearer stablecoins can be transferred directly or exchanged at market prices, whereas transfers between bank deposits are normally processed at face value through interbank settlement, typically using central bank reserves. Bank deposits can cease to exchange at par when the issuing bank fails and depositors face losses. This paper focuses on settlement costs and allows each issuer to choose the unit of account of its liabilities, so their effective denominations can coincide or differ in equilibrium. These choices interact with differences in the assets available for repayment and determine whether liabilities are valued at par, circulate across issuers, and support efficient trade.

Building on Doepke and Schneider (2017), this paper develops a three-period banking model in which banks choose the unit of account of their debt and whether that debt circulates as a transferable medium of exchange. The model can be interpreted through two USD stablecoins issued within the same monetary system. The stablecoins share the

same nominal denomination, while their effective denomination is the basket promised in repayment and may differ across issuers. Each issuer has a separate production basket, which represents the asset portfolio available to honour that promise. Two liabilities can therefore share a nominal denomination while differing in their effective denomination and state-contingent value. The model shows how these differences can weaken monetary singleness and how central bank money can counteract that effect. Reserves can restore efficient, fee-free exchange, while cash can support interoperability by providing an alternative settlement channel.

Making debt transferable across issuers creates a settlement obligation because the settling issuer must honour the repayment promised by the transferred claim using its own resources. Issuers have different asset bases, and their choices of effective denomination determine how the value of promised repayments moves relative to the resources available for settlement. Because the model excludes default, issuers must be able to honour promised repayments in every state. Allowing cross-issuer circulation may therefore tighten the promise-keeping constraint when the value of the transferred claim is high relative to the settling issuer's available resources. A tighter promise-keeping constraint reduces debt capacity and investment. A transfer fee reduces the quantity of liabilities transferred across issuers and therefore the amount requiring settlement. The fee also restricts circulation and reduces the value of the liability as a medium of exchange. If the cost of cross-issuer settlement becomes sufficiently large, it may be optimal for the issuer not to make its debt transferable across issuers.

A simple two-state version of the model makes the effects of differences between issuers' asset bases transparent. The analysis first considers what I call the natural unit of account, under which each issuer denominates its liabilities in its own production basket, which in the stablecoin example can be interpreted as its asset portfolio. Under this choice, changes in relative prices alter the value of the repayment promised by one issuer relative to the resources available to the other issuer for settlement. When relative-price variation is low, the resulting departures from par are small enough to be accommodated without constraining investment, and liabilities circulate across issuers without a transfer fee. As these departures become larger, the settlement constraint begins to reduce investment even though transfers remain fee-free. A further increase in the settlement burden makes positive transfer fees optimal, allowing cross-issuer circulation to continue while reducing the amount requiring settlement. If settlement becomes sufficiently costly, issuers prohibit cross-issuer transfers and the economy moves to autarky.

The two-state analysis also considers a common unit of account, under which both issuers promise repayment in the same basket. This common effective denomination ensures that their liabilities have the same state-contingent value and are therefore valued at

par. It does not necessarily align the promised repayment with the resources available to each issuer, so parity can coexist with constrained investment. The comparison between the two cases shows that exact parity is neither necessary nor sufficient for efficient exchange. Small departures from par can be consistent with efficient exchange, while larger departures eventually constrain investment.

A related question is how policy instruments affect singleness once private incentives create pressures towards fragmentation. A common-unit-of-account mandate removes settlement exposures created by differences in effective denomination, but can constrain investment when the mandated unit is poorly aligned with issuer resources. Central bank reserves provide a common settlement asset, while cash provides an alternative public settlement channel when direct transfers across issuers are costly, as discussed by Armelius, Claussen, and Hendry (2020), BIS (2003), and Rivadeneyra, Hendry, and García (2024). These instruments therefore affect settlement through different mechanisms.

The relationship between the unit of account and economic efficiency is developed by Doepke and Schneider (2017), who show how a common unit of account reduces relative-price risk along credit chains. Drenik, Kirpalani, and Perez (2022) relate preferences over units of account to the co-movement between currency values and consumption needs, while Zúñiga (2023) compares the transactional efficiency of alternative monetary regimes. This paper makes the choice of unit of account endogenous and links that choice to settlement, transferability, and the assets available to issuers for repayment. Bidder (2025) calls for greater care in assessing monetary singleness and questions the degree of concern attached to the small and noisy departures from par observed in normal times. The model illustrates the importance of the size of these departures. Limited departures can be consistent with efficient exchange when settlement constraints remain slack, whereas larger departures can constrain investment and circulation.

The paper also relates to work on short-term bank debt as a medium of exchange and to models of decentralised monetary exchange. Lagos and Wright (2005) provide a framework for studying how monetary arrangements affect exchange in decentralised markets. The model makes the use of liabilities in exchange endogenous by allowing issuers to choose whether their debt circulates and, if so, the terms on which it is transferred. Gorton and Pennacchi (1990) analyse how the information-insensitive design of demandable debt can support its circulation by limiting adverse selection. Gorton (2020) and Dang, Gorton, and Holmström (2020) emphasise the role of stable secondary-market values in allowing debt to circulate without counterparties investigating its quality. This literature links the circulation of debt to information about its quality and the risk of default. The baseline model studies a distinct mechanism in which settlement constraints can generate fragmentation under symmetric information and without default. An extension

introduces information sensitivity at the point of exchange and examines how the effective denomination of a liability affects its acceptance as a medium of exchange.

The paper also relates closely to recent work on digital money and payment-system fragmentation. Brunnermeier, James, and Landau (2021) study how platform currencies can displace national monies through digital dollarisation, while Ozdenoren, Tian, and Yuan (2025) examine how proprietary currencies allow platform issuers to earn seigniorage at the cost of greater fragmentation. Garratt and Shin (2023) compare bearer digital instruments with tokenised deposits and show how information sensitivity and doubts about backing assets can generate departures from par. Their mechanism is distinct from the symmetric-information settlement friction studied here. For stablecoins narrowly backed by transparent Treasury assets, backing uncertainty is likely to play a smaller role, and observed departures from par may also reflect limits to arbitrage or settlement frictions. Chiu and Monnet (2025) show how differences in the programmability and transferability of digital currencies create a trade-off between commitment and information frictions. Huang and Keister (2026) compare stablecoins backed by safe assets with bank deposits used to finance broader asset portfolios, and show how their relative benefits depend on bank regulation and incentives to take risk. This paper complements their analysis by examining how the interaction between the effective denomination of a liability and the assets available for repayment affects settlement and the circulation of private money.

Section 2 presents the model. Section 3 characterises equilibrium and derives conditions for efficient interbank trade. Section 4 studies the two-state case and illustrates the equilibrium structure with numerical examples. Section 5 analyses three policy instruments, namely a common-unit-of-account mandate, central bank reserves, and cash, and characterises their interaction. Section 6 studies how informational frictions at the point of exchange interact with the baseline mechanism. Section 7 concludes.

2. The Model

The model has two banks, each of which produces a distinct basket of goods and issues debt to labourers. Banks choose the basket promised in repayment, which determines the effective denomination of their liabilities. This promised basket is distinct from the bank's production basket, which represents the resources available for repayment. The banks represent issuers of private liabilities and admit several interpretations, including two issuers of USD stablecoins. Although both liabilities are denominated in dollars, their issuers may choose different repayment baskets, giving the liabilities different effective denominations and potentially different realised values across states.

Banks also choose whether their debt can be transferred across regions as a medium of exchange. A cross-region transfer may require the issuing bank to settle a claim whose value moves differently from the value of its production resources. Meeting this obligation can reduce the amount of debt the bank can issue and affect the terms on which its liabilities circulate.

2.1 Environment

There are two goods, $i \in \{A, B\}$, each of which serves as both a production output and a consumption good. Time is discrete, with three dates, 0, 1, and 2.

There are two regions. Each region contains three agents: a bank, a labourer, and an artisan. At date 0, the bank and labourer in the same region enter into a bilateral contract. The labourer provides labour input n to the bank in exchange for a debt contract, which is described below. At date 1, the labourer may meet an artisan. Labourers can be interpreted as money holders, artisans as merchants, and meetings as payment opportunities. If a meeting occurs, the artisan can expend x units of labour effort to produce x units of a customised good. This good can be consumed only by the labourer for whom it is produced at date 2. It cannot be traded on the open market and has no value to other agents.

At date 2, the bank's output is realised and a spot market opens in which goods A and B are traded. As in Doepke and Schneider (2017), the price vector $\mathbf{p}$ is exogenous and random, with compact support whose convex hull is denoted by $\mathbf{P} \subset \mathbb{R}^2_{>0}$. The price process captures future variation in the relative value of the two goods that banks take as given when choosing their contracts. I assume that there exists a vector $\underline{\mathbf{p}} \gg \mathbf{0}$ such that $\mathbf{p} \geq \underline{\mathbf{p}}$ for every $\mathbf{p} \in \mathbf{P}$. Prices are therefore bounded away from zero in every state.

All agents receive instantaneous utility $v(\mathbf{c})$ from consuming a vector $\mathbf{c}$ of goods A and B at date 2, where $v(\cdot)$ is homogeneous of degree 1. The bank derives utility only from consumption of tradable goods at date 2.

With probability $\alpha > 0$, the labourer meets an artisan in the same region. With probability $\beta > 0$, the labourer meets an artisan in the other region. With probability $1 - \alpha - \beta > 0$, no meeting occurs. These probabilities are symmetric across regions. An artisan who expends x units of labour to produce a customised good and then consumes a vector $\mathbf{c}$ of tradable goods has utility

$$v(\mathbf{c}) - x.$$

A labourer who provides n units of labour at date 0, receives x units of the customised good, and consumes a vector $\mathbf{c}$ of tradable goods at date 2 has utility

$$v(\mathbf{c}) + (1+\gamma)x - n,$$

where $\gamma > 0$ implies gains from trade between the labourer and the artisan. Labourers cannot issue debt of their own. Trade between labourers and artisans can therefore occur only if bank debt can be transferred and used as a means of payment.

Each bank produces a non-negative vector of goods denoted by $\mathbf{y}_j$ according to the following production function:

$$\mathbf{y}_j = n_j^{\varphi} \hat{\mathbf{y}}_j, \tag{1}$$

where n_j is the scalar quantity of labour supplied to bank j by the labourer located in the same region. The parameter $\varphi \in (0,1)$ gives diminishing returns to labour. Each bank produces goods A and B in a fixed ratio determined by the unit vector $\hat{\mathbf{y}}_j$, normalised so that $\|\hat{\mathbf{y}}_j\|_1 = 1$, which I refer to as bank j's unit of production. Each element of $\hat{\mathbf{y}}_j$ is weakly positive, so production of each good is non-negative. In the stablecoin example, I interpret the unit of production as the composition of the issuer's asset portfolio and hence the resources available for repayment. It is distinct from the basket defining the repayment promised by the liability, which is introduced below.

As in Doepke and Schneider (2017), I impose two price normalisations. First, the price vector $\mathbf{p}$ satisfies $\tilde{P}(\mathbf{p}) = 1$ for every $\mathbf{p} \in \mathbf{P}$, where $\tilde{P}(\mathbf{p})$ is the expenditure function required to attain one unit of utility:

$$\tilde{P}(\mathbf{p}) = \min_{\mathbf{c}} \{\mathbf{p}'\mathbf{c}\} \quad \text{subject to} \quad v(\mathbf{c}) \geq 1.$$

Together with the assumption that v is homogeneous of degree one, this normalisation implies that the indirect utility obtained from the proceeds of a bundle $\mathbf{y}$ of goods A and B is $\mathbf{p}'\mathbf{y}$. Second, the expected price of each good is normalised to one:

$$\mathbb{E}[p^i] = 1, \qquad i \in \{A, B\}.$$

2.2 Contracting Frictions

A key assumption is that the relative-price risk between the two banks' units of production cannot be insured at date 0. If complete state-contingent insurance were available, each bank could hedge changes in the value of its resources relative to the repayment promised by its debt. The choice of unit of account would then have no effect on feasibility. In

the absence of such insurance, promised repayment and the resources available to meet it may move differently as relative prices change.

The relevant missing market is for contracts that insure these future relative values. The assumption captures the practical difficulty of writing contracts at date 0 that span every future realisation of the relevant relative prices. Liquid and readily valued underlying assets do not themselves provide such insurance, since their liquidity does not eliminate variation in their value relative to the repayment promised by the liability. The assumption requires only that banks cannot contract at date 0 to eliminate every such future mismatch.

A second assumption is that labourers cannot issue debt of their own. Trade with artisans must therefore be financed using debt issued by a bank, making bank debt the unique means of payment in these meetings. The bank chooses the debt contract subject to the labourer's participation constraint and the bank's feasibility constraints. These constraints determine how much debt the bank can issue and hence the quantity available for use in exchange.

For bank debt to serve as a means of payment, the labourer must be able to transfer it to an artisan. The debt contract specifies whether the labourer can transfer the claim beyond the issuing region and the terms on which any transfer occurs. Non-transferable debt remains a bilateral claim between the bank and the labourer. Transferable debt can be accepted by an artisan in exchange for the customised good, allowing the claim to circulate beyond the original contract.

A third assumption is that banks must honour their debt obligations in every state, so default does not occur. Each debt contract is subject to a promise-keeping constraint requiring the bank to have sufficient resources to meet the promised repayment for every price vector in $\mathbf{P}$. When the value of the promised repayment varies relative to the value of the bank's production, this constraint may limit the amount of debt that the bank can issue.

For each bank, repayment feasibility is governed by its own worst-case price realisation, namely the realisation in which the promised repayment has its highest value relative to the resources available to meet it. The worst-case realisation may differ across banks because their units of production and repayment promises may differ. Debt can be transferred across regions only if the resulting obligations remain feasible for both banks in every state. The settlement requirement introduced below formalises this state-by-state comparison for each bank.

The contracting environment assumes that each bank uses a single unit of account for debt issued to labourers, which may subsequently be accepted by artisans. This captures

the use of a common denomination for liabilities offered to retail customers. Banks are assumed to have greater flexibility when issuing claims to one another for settlement, including the ability to choose a unit of account suited to a particular interbank obligation. This flexibility does not alter the repayment promised by the underlying retail debt or insure every future relative-price realisation. Together with the absence of complete date-0 insurance, this distinction between retail and interbank contracting means that differences in the units of account chosen for bank debt continue to affect settlement feasibility.

2.3 Bank Debt

At date 0, bank j makes a take-it-or-leave-it offer to the labourer in its region. The labourer provides n_j units of labour in exchange for bank debt $\mathbf{b}_j$, which is a promise to deliver a bundle of goods at date 2. I decompose this repayment promise into its scale, $B_j > 0$, and its unit of account, $\hat{\mathbf{b}}_j$, so that $\mathbf{b}_j = B_j \hat{\mathbf{b}}_j$, where

$$\hat{\mathbf{b}}_j = \frac{\mathbf{b}_j}{\|\mathbf{b}_j\|_1}. \tag{2}$$

The scalar B_j determines the size of the repayment promise, while the unit vector $\hat{\mathbf{b}}_j$ determines its composition across goods A and B. For a given price realisation $\mathbf{p}$, one unit of the liability has value $\mathbf{p}'\hat{\mathbf{b}}_j$. In the stablecoin example, I interpret $\hat{\mathbf{b}}_j$ as the liability's effective denomination, meaning the basket promised in repayment. This repayment basket is distinct from the bank's unit of production $\hat{\mathbf{y}}_j$, which represents the resources available to honour the promise.

The contract also specifies whether the debt can be transferred to another agent. If it is transferable, it can be used as a means of payment and can therefore facilitate trade between labourers and artisans. Transferability may be restricted to artisans in the same region or extended across regions. The bank may also charge a fee on transfers. The fee $\chi_{j,k} \geq 0$ is the fraction of bank j's debt paid to bank j when a labourer transfers the debt to an agent in region k. The recipient therefore receives the transferred debt net of this fee. The fee may differ between transfers within the same region and transfers across regions. The offer is subject to the labourer's participation constraint and the feasibility constraints set out below, and banks commit fully to the terms agreed at date 0. In this and the other take-it-or-leave-it offers in the model, a recipient who is indifferent between accepting and rejecting an offer accepts it.

At date 2, the bank must honour the repayment promised to the labourer for every possible price realisation $\mathbf{p}$. Debt that has not been transferred is therefore subject to

the following promise-keeping constraint:

$$\mathbf{p}' \left(n_j^{\varphi} \hat{\mathbf{y}}_j - B_j \hat{\mathbf{b}}_j \right) \geq 0 \quad \forall \mathbf{p} \in \mathbf{P}. \tag{3}$$

The constraint requires the value of bank j's production to cover the value of its promised repayment in every state.

At date 1, a labourer may meet an artisan and attempt to trade. If bank j's debt is transferable, the labourer can request that bank j transfer a fraction $s_{j,k}$ of the debt to an artisan in region k. When the artisan is located in region j, the transferred debt retains the same unit of account as the debt originally issued to the labourer. An intrabank transfer therefore creates no additional settlement obligation arising from a change in denomination. Since $\gamma > 0$, such transfers generate gains from trade whenever a meeting occurs. An intrabank transfer fee $\chi_{j,j}$ would reduce these gains without relaxing the promise-keeping constraint in equation (3), which must also hold when no meeting occurs. Bank j therefore optimally sets $\chi_{j,j} = 0$.

2.4 Interbank Settlement

If the labourer requests that bank j transfer a fraction $s_{j,k} \in [0, 1]$ of its debt to an artisan in region $k \neq j$, bank k must agree to facilitate the transfer. Bank j, which initiates the arrangement, makes a take-it-or-leave-it offer to bank k. The offer specifies the amount of bank j's debt to be transferred, $s_{j,k} B_j \hat{\mathbf{b}}_j$, and an interbank claim $\mathbf{t}_{j,k}$ issued by bank j to bank k in settlement. This interbank claim promises a bundle of tradable goods and can be decomposed into its scale, T_j, and unit of account, $\hat{\mathbf{t}}_{j,k}$, so that

$$\mathbf{t}_{j,k} = T_j \hat{\mathbf{t}}_{j,k},$$

where $\hat{\mathbf{t}}_{j,k}$ is a unit vector.

If bank k accepts the offer, it receives the interbank claim $T_j \hat{\mathbf{t}}_{j,k}$ from bank j and facilitates the transfer to the artisan in region k. The fraction $s_{j,k}$ of bank j's debt requested for transfer is reduced by the fee $\chi_{j,k}$. Bank k then issues debt to the artisan in its own unit of account, with a value equal to the net amount transferred after the fee.

Bank k uses the same unit of account for debt issued to labourers and artisans within its region. The transferred claim must therefore be converted from bank j's unit of account into bank k's unit of account. At date 1, this conversion is based on the ratio of the

expected values of the two liabilities:

$$\frac{\mathbb{E}\left[\mathbf{p}'\hat{\mathbf{b}}_j\right]}{\mathbb{E}\left[\mathbf{p}'\hat{\mathbf{b}}_k\right]}.$$

The expected-price normalisation $\mathbb{E}[p^i] = 1$ for each $i \in \{A, B\}$, together with the restriction that $\hat{\mathbf{b}}_j$ and $\hat{\mathbf{b}}_k$ are unit vectors, implies that

$$\mathbb{E}\left[\mathbf{p}'\hat{\mathbf{b}}_j\right] = \mathbb{E}\left[\mathbf{p}'\hat{\mathbf{b}}_k\right] = 1.$$

The date-1 conversion rate is therefore equal to one. This conversion rate determines the quantity of bank k's debt issued against the transferred claim, while the transfer fee $\chi_{j,k}$ separately reduces the amount passed through to the artisan.

Bank k must be able to honour the obligation issued to the artisan for every possible price realisation $\mathbf{p}$. I assume that the interbank claim received from bank j is sufficient to cover this obligation. Bank k can therefore accept the transfer only if

$$\mathbf{p}'\left(T_j\hat{\mathbf{t}}_{j,k} - s_{j,k}\left(1-\chi_{j,k}\right)B_j\hat{\mathbf{b}}_k\right) \geq 0 \quad \forall \mathbf{p} \in \mathbf{P}. \tag{4}$$

The transfer also changes bank j's obligations. The fraction $s_{j,k}$ of its debt transferred by the labourer is cancelled, leaving bank j with the remaining obligation $(1-s_{j,k})B_j\hat{\mathbf{b}}_j$. Bank j also issues the interbank claim $T_j\hat{\mathbf{t}}_{j,k}$ to bank k. The transfer is feasible for bank j only if its output is sufficient to meet both obligations for every possible price realisation:

$$\mathbf{p}'\left(n_j^{\varphi}\hat{\mathbf{y}}_j - \left(1-s_{j,k}\right)B_j\hat{\mathbf{b}}_j - T_j\hat{\mathbf{t}}_{j,k}\right) \geq 0 \quad \forall \mathbf{p} \in \mathbf{P}. \tag{5}$$

The bank that initiates an interbank transfer has all the bargaining power in its negotiation with the accepting bank. In the transfer considered here, bank j initiates the transfer and therefore chooses the interbank claim so that bank k earns zero profit from accepting it in every price realisation. Bank k's promise-keeping constraint in equation (4) then binds state by state:

$$T_j\mathbf{p}'\hat{\mathbf{t}}_{j,k} = s_{j,k}(1-\chi_{j,k})B_j\mathbf{p}'\hat{\mathbf{b}}_k \quad \forall \mathbf{p} \in \mathbf{P}. \tag{6}$$

Bank j can ensure that this equality holds for every price realisation by writing the interbank claim in the same unit of account as the debt that bank k issues to the artisan. Hence,

$$\hat{\mathbf{t}}_{j,k} = \hat{\mathbf{b}}_k.$$

The scale T_j is then chosen so that bank k's promise-keeping constraint binds in every state.

This contract ensures that the interbank claim received by bank k exactly covers its obligation to the artisan in every state. Bank j issues an interbank obligation denominated in bank k's unit of account and must honour that obligation using its own production. Substituting the optimal interbank claim into bank j's promise-keeping constraint in equation (5) and evaluating it at the full-transfer endpoint, $s_{j,k} = 1$, gives

$$\mathbf{p}' \left(n_j^{\varphi} \hat{\mathbf{y}}_j - (1 - \chi_{j,k}) \, \hat{\mathbf{b}}_k B_j \right) \geq 0 \quad \forall \mathbf{p} \in \mathbf{P}. \tag{7}$$

The corresponding condition at $s_{j,k} = 0$ is equation (3). Since the promise-keeping condition is affine in $s_{j,k}$, satisfying these two endpoint conditions ensures feasibility for every transfer fraction $s_{j,k} \in [0, 1]$.

Equation (7) identifies the state-contingent exposure bank j faces following a cross-region transfer. The interbank obligation is denominated in bank k's unit of account, so its value in state $\mathbf{p}$ is determined by $\mathbf{p}'\hat{\mathbf{b}}_k$. Bank j must honour this obligation using production with value $\mathbf{p}'\hat{\mathbf{y}}_j$. Changes in relative prices may therefore alter the value of the promised repayment relative to the resources available to meet it. When this state-contingent exposure tightens bank j's promise-keeping constraint, it reduces the amount of debt that the bank can issue.

The transfer fee $\chi_{j,k}$ reduces the repayment obligation created by the transfer through the term $(1-\chi_{j,k})B_j\hat{\mathbf{b}}_k$. A higher fee can therefore relax bank j's promise-keeping constraint by reducing the amount that must be settled across banks. The fee also reduces the amount received by the artisan and is therefore limited by the labourer's participation constraint. As shown below, when the settlement exposure is sufficiently large, the bank cannot use the transfer fee to prevent its debt capacity from falling.

Bank j must be able to settle the interbank obligation for every price realisation. The relevant requirement is therefore determined by the price realisation in which the obligation is most costly relative to bank j's production:

$$\epsilon_{j,k} \equiv \max_{\mathbf{p} \in \mathbf{P}} \frac{\mathbf{p}'\hat{\mathbf{b}}_k}{\mathbf{p}'\hat{\mathbf{y}}_j}. \tag{8}$$

For each unit of debt denominated in bank k's unit of account, $\epsilon_{j,k}$ measures the largest settlement obligation relative to the value of bank j's production across price realisations. The ratio is continuous on the compact set $\mathbf{P}$, and its denominator is bounded away from zero. The maximum is therefore finite and attained.

The expected-price normalisation and the unit-vector restrictions imply that $\epsilon_{j,k} \geq 1$. The settlement requirement equals one when $\hat{\mathbf{b}}_k = \hat{\mathbf{y}}_j$. It is strictly greater than one whenever the promised repayment is more valuable than bank j's unit of production for some price realisation. When $\epsilon_{j,k} > 1$, the excess $\epsilon_{j,k} - 1$ measures the additional resources required in bank j's worst-case price realisation. The results below show how this settlement exposure reduces debt capacity and changes the terms on which the bank is willing to make its debt transferable across regions.

Substituting the relevant settlement requirements into the promise-keeping constraints yields two reduced conditions. For debt that remains within bank j, feasibility requires

$$n_j^{\varphi} \frac{1}{\epsilon_{j,j}} - B_j \geq 0, \tag{9}$$

where $\epsilon_{j,j}$ measures the value of repayment promised in bank j's own unit of account relative to the value of its production in the worst-case price realisation. This constraint limits the debt capacity of bank j before any cross-region transfer.

If bank j's debt is transferred to region k, the corresponding promise-keeping constraint is

$$n_j^{\varphi} \frac{1}{\epsilon_{j,k}} - (1 - \chi_{j,k})\, B_j \geq 0. \tag{10}$$

The settlement requirement $\epsilon_{j,k}$ reflects the value of an obligation denominated in bank k's unit of account relative to bank j's production. When this requirement exceeds one, settlement reduces the amount of transferable debt that bank j can support for a given level of production. A higher transfer fee relaxes the constraint by reducing the amount passed through to the artisan, although the labourer's participation constraint limits the bank's use of this margin.

2.5 Decentralised Trade

Transferability allows bank debt to circulate as a means of payment and creates gains from trade between labourers and artisans. When debt is transferred across banks, these gains must be weighed against the settlement requirement derived above. Transferability can only be sustained when the gains from trade outweigh the associated settlement costs.

Consider a labourer in region j who provides n_j units of labour to bank j in exchange for debt $\mathbf{b}_j = B_j \hat{\mathbf{b}}_j$. If the labourer does not meet an artisan at date 1, the labourer retains the debt and receives the promised bundle of tradable goods at date 2. Given the price normalisation and the homogeneity of $v(\cdot)$, the labourer's expected utility is

$$\mathbb{E}[\mathbf{p}'\mathbf{b}_j] - n_j = B_j - n_j. \tag{11}$$

The same expression gives the labourer's expected utility when debt is non-transferable, since the debt cannot then be used to trade with an artisan at date 1.

Now suppose that the labourer in region j meets an artisan in region k. If bank j's debt is transferable, the labourer can use it to purchase the customised good, subject to the transfer fee $\chi_{j,k}$ agreed at date 0. The labourer has all the bargaining power and makes a take-it-or-leave-it offer to the artisan. The offer specifies a fraction $s_{j,k} \in [0,1]$ of the debt to be transferred in exchange for x_k units of the customised good. The labourer chooses $s_{j,k}$ and x_k to maximise

$$\max_{s_{j,k},x_k} (1 - s_{j,k})\, \mathbb{E}[\mathbf{p}'\mathbf{b}_j] + (1+\gamma)\, x_k - n_j, \tag{12}$$

subject to the artisan's participation constraint

$$s_{j,k}\,(1 - \chi_{j,k})\, \mathbb{E}[\mathbf{p}'\mathbf{b}_j] - x_k \geq 0. \tag{13}$$

The labourer's problem has a corner solution. If trade occurs, the labourer transfers the full debt claim, so $s_{j,k} = 1$ and chooses x_k so that the artisan's participation constraint binds. Otherwise, the labourer sets $s_{j,k} = x_k = 0$ and retains the debt.

Since $\gamma > 0$, the labourer trades whenever the transfer fee leaves sufficient value to compensate the artisan. With the artisan's participation constraint binding, trade gives the labourer $(1+\gamma)(1-\chi_{j,k})$ units of utility for each unit of debt transferred, rather than the one unit obtained by retaining the debt. Trade therefore occurs when

$$(1+\gamma)\,(1 - \chi_{j,k}) \geq 1. \tag{14}$$

Equivalently, the transfer fee cannot exceed $\gamma/(1+\gamma)$ if the debt is to be used in exchange. At equality, the labourer is indifferent between trading and retaining the debt, and I assume that the labourer then transfers the full claim.

Suppose that the transfer fee is low enough for trade to occur. The labourer's expected utility from tradable debt is

$$\begin{aligned} U_j =& \alpha\,(1+\gamma)\,(1-\chi_{j,j})\,\mathbb{E}[\mathbf{p}'\mathbf{b}_j] \\ &+ \beta\,(1+\gamma)\,(1-\chi_{j,k})\,\mathbb{E}[\mathbf{p}'\mathbf{b}_j] \\ &+ (1-\alpha-\beta)\,\mathbb{E}[\mathbf{p}'\mathbf{b}_j] - n_j. \end{aligned} \tag{15}$$

The first term applies when the labourer meets an artisan in the same region, while the second applies when the artisan is located in the other region. The final term applies when no meeting occurs.

Using the expected-price normalisation, this expression becomes

$$U_j = \left[(1+\gamma)\left(\alpha\left(1-\chi_{j,j}\right)+\beta\left(1-\chi_{j,k}\right)\right)+\left(1-\alpha-\beta\right)\right]B_j - n_j. \tag{16}$$

Equation (16) determines the labourer's participation constraint in the bank's contract problem. Transferability increases the value of bank debt to the labourer when it permits additional trade, while transfer fees reduce this benefit. The labourer may therefore be willing to supply more labour when debt is transferable and transfer fees are sufficiently low. Since labourers are risk neutral and expected claim values are normalised, the unit of account does not enter their expected utility directly. The unit of account affects the contract only indirectly through the bank's promise-keeping constraints.

Transferability increases the value of bank debt to labourers by allowing it to be used in exchange. For a given labour input, this allows the bank to offer a smaller repayment promise while still satisfying the labourer's participation constraint, increasing the bank's capacity to finance production. Transfers across banks create additional gains from trade when the labourer meets an artisan in the other region, but also expose the issuing bank to the settlement requirement $\epsilon_{j,k}$. A transfer fee reduces this exposure by lowering the amount requiring settlement, while also reducing the benefit that transferability provides to the labourer. The bank's contract problem therefore determines the transfer fee and debt issuance by balancing the gains from exchange against the associated settlement burden.

2.6 Timeline

At date 0, bank j makes a take-it-or-leave-it offer to the labourer in its region. The contract specifies the labour input n_j, the promised repayment $\mathbf{b}_j$, its unit of account, and whether the debt is transferable. It also specifies any fee charged when the debt is transferred. If the labourer accepts the offer, the bank receives the labour input used in production and the labourer becomes a debtholder.

At date 1, labourers may meet artisans. A labourer who holds transferable debt can make a take-it-or-leave-it offer to an artisan, specifying the fraction of bank debt to be transferred in exchange for a quantity x of the customised good. A transfer within the same region is completed by the issuing bank. For a transfer to an artisan in another region, the issuing bank makes a take-it-or-leave-it offer to the bank in that region. The initiating bank issues an interbank claim to the accepting bank, which in turn issues debt to the artisan in its own unit of account.

Table 1: Key model notation

Model object	Formal role	Interpretation
B_j	Scale of the repayment promise	Size of the liability issued by bank j
$\hat{\mathbf{b}}_j$	Unit of account of bank j's debt	Composition of the basket promised per unit of the liability; interpreted as its effective denomination in the stablecoin example
$\mathbf{b}_j = B_j\hat{\mathbf{b}}_j$	Date-2 bundle promised by bank j	Total repayment promised by the liability
$\hat{\mathbf{y}}_j$	Unit of production of bank j	Composition of the bank's output; interpreted as the issuer's asset portfolio and resources available for repayment in the stablecoin example
$\chi_{j,k}$	Fee charged when bank j's debt is transferred to region k	Fraction of the transferred debt retained by bank j, reducing the amount passed through to the artisan
$\epsilon_{j,k}$	Worst-case settlement requirement faced by bank j	Largest value of an obligation denominated in bank k's unit of account relative to the value of bank j's production across price realisations

At date 2, prices and bank production are realised and a competitive spot market opens for goods A and B. Banks use the resources available after spot-market trade to honour the repayments promised by their debt and interbank claims.

2.7 Constrained-Efficient Allocation

The preceding analysis shows that transferability creates gains from trade but may also tighten banks' promise-keeping constraints. To assess whether the resulting equilibrium allocation is efficient, I compare it with a constrained-efficient benchmark. The benchmark retains the banks' production technology and each bank's take-it-or-leave-it debt contract with the labourer in its region, but abstracts from the promise-keeping constraints in equations (9) and (10). An equilibrium is inefficient when the allocation does not attain this benchmark.

Under the benchmark, transfer fees reduce the value of debt to the labourer by more than they reduce the bank's repayment cost, so banks set them to zero and trade takes place whenever a labourer meets an artisan. One unit of bank debt yields a marginal benefit of 1 when no meeting occurs and $1+\gamma$ when a meeting creates an additional gain from trade. Since a meeting occurs with probability $\alpha+\beta$, the expected value of debt to the labourer is $H \equiv 1+\gamma(\alpha+\beta)$. Bank j's profit falls with B_j, so its take-it-or-leave-it offer makes the labourer's participation constraint from equation (16), $HB_j \geq n_j$, bind, giving $B_j = n_j/H$. Since the labourer and the artisan then receive no surplus, bank profit equals total surplus under the benchmark. Banks' choices therefore give

$$\mathcal{W} = \max_{\{n_j\}_j} \sum_j \left[\mathbb{E}[\mathbf{p}'\hat{\mathbf{y}}_j] n_j^{\varphi} - \frac{n_j}{H} \right], \tag{17}$$

where the expected-price normalisation implies $\mathbb{E}\left[\mathbf{p}'\hat{\mathbf{y}}_j\right] = 1$.

Each bank chooses labour input by equating the marginal benefit of an additional unit of labour in production to its marginal cost. Since the expected value of each bank's unit of production is one, the optimal labour input is the same across banks and is given by

$$n_j^* = \left[\varphi\left(1+\gamma\left(\alpha+\beta\right)\right)\right]^{\frac{1}{1-\varphi}}. \tag{18}$$

Substituting this choice into the objective gives the benchmark value:

$$\mathcal{W} = \sum_j \left(1-\varphi\right)\left[\varphi\left(1+\gamma\left(\alpha+\beta\right)\right)\right]^{\frac{\varphi}{1-\varphi}}. \tag{19}$$

The per-bank benchmark value $W^* \equiv \mathcal{W}/2$ increases with the likelihood of trade, measured by the meeting probabilities α and β, and with the gains from trade, γ.

3. Equilibrium

This section characterises equilibrium in three steps. It first studies the contract a bank offers when liabilities circulate only within its own region. It then analyses the contract when cross-bank transfers are allowed and shows how banks decide whether to permit them. Finally, it characterises strategic interaction between banks over units of account, transfer fees, and whether to allow interbank transfers.

Within a given contract, bank j chooses the labour input n_j, the size of debt B_j, the unit of account in which that debt is denominated, and the conditions under which the debt can be transferred. Because transfers within the same bank create no additional settlement obligation, the central contractual choice is whether debt can be transferred across banks and, if so, the interbank transfer fee $\chi_{j,k}$ that applies.

3.1 Contracting under Autarky

First, consider the contract bank j offers when it does not allow interbank transfers. In that case, its liabilities circulate only within its own region. Bank j chooses labour n_j, debt size B_j, its unit of account, and the intrabank transfer fee $\chi_{j,j}$ to maximise expected profit conditional on not allowing interbank transfers.

The bank's optimisation problem is to maximise its profit

$$V_j = \max_{\left\{n_j, B_j, \hat{\mathbf{b}}_j, \chi_{j,j}\right\}} n_j^{\varphi} - \left(1 - \alpha\chi_{j,j}\right) B_j, \tag{20}$$

subject to the participation constraint of the labourer, the promise-keeping constraint on its debt, and a maximum intrabank transfer fee that would be paid by the labourer.

The participation constraint requires the labourer's expected utility to be non-negative, so $U_j \geq 0$. Restricting interbank transfers eliminates gains from trade with foreign artisans. This yields

$$\left[\alpha\left(1+\gamma\right)\left(1-\chi_{j,j}\right) + 1 - \alpha\right] B_j \geq n_j. \tag{21}$$

The bank must ensure that the promise-keeping constraint, equation (9), holds in all states. This constraint also covers the case where the labourer transfers the debt to an artisan within the same region, regardless of whether an intrabank fee is imposed.

There is an upper limit on the intrabank transfer fee. Since the gain from trade between labourers and artisans is $(1+\gamma)$, the fee must not be large enough to eliminate this benefit entirely. The upper bound on $\chi_{j,j}$ is

$$(1+\gamma)(1-\chi_{j,j}) \geq 1. \tag{22}$$

The following proposition characterises the optimal contract when bank j does not allow interbank transfers.

Proposition 1 (Autarky Contract)**.** *Suppose bank j does not allow interbank transfers. Then the contract under autarky features zero intrabank fees ($\chi_{j,j} = 0$), investment*

$$n_j^{(0)} = [\varphi(1+\gamma\alpha)]^{\frac{1}{1-\varphi}},$$

debt size $B_j^{(0)} = \varphi^{\frac{1}{1-\varphi}}(1+\gamma\alpha)^{\frac{\varphi}{1-\varphi}}$, profit $V_j^{(0)} = \varphi^{\frac{\varphi}{1-\varphi}}(1+\gamma\alpha)^{\frac{\varphi}{1-\varphi}}(1-\varphi)$, and any unit of account satisfying $\epsilon_{j,j} \leq 1/\varphi$.

Proof. See Appendix A. □

It is optimal to set the intrabank fee to zero, so $\chi_{j,j} = 0$. Raising the fee increases fee revenue, but it also reduces the amount of debt the labourer can transfer to the artisan by the same amount. Since $\gamma > 0$, the gains from trade are positive. As a result, a higher fee raises the bank's cost of debt by more than it raises fee revenue.

Both the labour input chosen by the bank, $n_j^{(0)}$, and the profit it earns, $V_j^{(0)}$, are below the constrained optimum since $\beta > 0$. When interbank transfers are not allowed, labourers cannot use bank debt to trade with foreign artisans. They therefore value the debt less, and the bank's borrowing capacity is lower than it otherwise would be.

In equilibrium, the labourer's participation constraint, equation (21), binds. Bank j's choice of unit of account enters the bank's problem through the settlement requirement $\epsilon_{j,j}$ in the promise-keeping constraint, equation (9). It is optimal for bank j to choose a unit of account that leaves this constraint slack. One such choice is the natural unit of account, $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$, which gives $\epsilon_{j,j} = 1$. The promise-keeping constraint then becomes $n_j^{\varphi} \geq B_j$ and is satisfied whenever the bank earns non-negative profit. More generally, any unit of account satisfying $\epsilon_{j,j} \leq 1/\varphi$ allows the optimal autarkic contract to be implemented, so the natural unit is not the only optimal choice. The promise-keeping constraint is slack when $\epsilon_{j,j} < 1/\varphi$ and holds with equality at $\epsilon_{j,j} = 1/\varphi$.

3.2 Contracting with Transferable Debt

Bank j maximises profit by choosing labour n_j, debt size B_j, its unit of account $\hat{\mathbf{b}}_j$, an intrabank fee $\chi_{j,j} \geq 0$, and an interbank fee $\chi_{j,k} \geq 0$.

$$W_j = \max_{\left\{\chi_{j,j},\chi_{j,k},\hat{\mathbf{b}}_j,B_j,n_j\right\}} n_j^{\varphi} - (1 - \alpha\chi_{j,j} - \beta\chi_{j,k})\, B_j, \tag{23}$$

subject to the labourer's participation constraint, the promise-keeping constraint on debt retained within region j, the upper bounds on transfer fees, and an additional promise-keeping constraint if the debt is transferred to region k.

The participation constraint of the labourer adds the gains from trade with foreign artisans to the autarky case:

$$\left[(1+\gamma)\left(\alpha\left(1-\chi_{j,j}\right)+\beta\left(1-\chi_{j,k}\right)\right)+(1-\alpha-\beta)\right] B_j \geq n_j. \tag{24}$$

The promise-keeping constraint on the original debt contract is the same as in the autarky case and is given by equation (9). There is also a second promise-keeping constraint on the interbank obligation, given by equation (10). The associated settlement requirement $\epsilon_{j,k}$ compares the value of the obligation denominated in bank k's unit of account with the value of bank j's production in the worst-case price realisation.

Finally, there are upper bounds on the transfer fees. The upper bound on the fee for an intrabank transfer is identical to that in the autarky case and is given by equation (22). The analogous upper bound on the interbank transfer fee is

$$(1+\gamma)\left(1-\chi_{j,k}\right) \geq 1. \tag{25}$$

Proposition 2 (Transferable Debt)**.** *Suppose bank j allows interbank transfers and bank k's unit of account induces state-contingent settlement requirement $\epsilon_{j,k}$. Then $\chi_{j,j} = 0$ and any unit of account with $\epsilon_{j,j} \leq n_j^{\varphi}/B_j$, including the natural unit of account, implements the contract, where n_j^{φ}/B_j equals $1/\varphi$ in case (i) and $\epsilon_{j,k}(1-\chi_{j,k}^*) \geq 1$ in case (ii):*

(i) If $\epsilon_{j,k} \leq 1/\varphi$, the interbank constraint does not reduce investment. Bank j sets $\chi_{j,k} = 0$ and achieves the constrained optimum with investment $n_j^{(1)} = n_j^*$, debt size $B_j^{(1)} = \varphi^{\frac{1}{1-\varphi}}(1+\gamma(\alpha+\beta))^{\frac{\varphi}{1-\varphi}}$, and profit

$$W_j^{(1)} = \varphi^{\frac{\varphi}{1-\varphi}}(1+\gamma(\alpha+\beta))^{\frac{\varphi}{1-\varphi}}(1-\varphi).$$

(ii) If $\epsilon_{j,k} > 1/\varphi$, the interbank constraint binds. Bank j sets $\chi_{j,k} = \chi^*_{j,k}(\epsilon_{j,k}) \geq 0$ (characterised in Appendix C), invests below the optimum ($n_j^{(2)} < n_j^{(1)} = n_j^*$), and earns profit

$$W_j^{(2)} = \left[\frac{1+\gamma\alpha+\beta\left(\gamma-(1+\gamma)\chi^*_{j,k}\right)}{\epsilon_{j,k}\left(1-\chi^*_{j,k}\right)}\right]^{\frac{\varphi}{1-\varphi}}\left(1-\frac{1-\beta\chi^*_{j,k}}{\epsilon_{j,k}\left(1-\chi^*_{j,k}\right)}\right),$$

which is strictly decreasing in $\epsilon_{j,k}$. Explicit expressions for $n_j^{(2)}$ and $B_j^{(2)}$ are given in Appendix B.

Proof. See Appendix B. □

Case (i) describes the benchmark in which the interbank promise-keeping constraint does not reduce investment. When $\epsilon_{j,k} \leq 1/\varphi$, bank j can satisfy the interbank obligation without reducing investment below the constrained optimum. Bank j therefore hires the optimal amount of labour, $n_j = n_j^*$, and allows its debt to be transferred to region k without charging an interbank fee, so $\chi_{j,k} = 0$.

In case (ii), bank k's unit of account induces a sufficiently large state-contingent settlement requirement, $\epsilon_{j,k} > 1/\varphi$. In the worst-case price realisation for bank j, the obligation denominated in $\hat{\mathbf{b}}_k$ is large relative to the value of bank j's production. The interbank promise-keeping constraint, equation (10), therefore binds, and it is no longer feasible to implement the constrained-optimal debt contract.

Bank j responds by reducing investment below n_j^*. When the settlement requirement exceeds the zero-fee cutoff derived in Appendix C, the bank also charges a positive interbank transfer fee. The fee allows bank j to reduce the amount requiring settlement, but it also reduces the value of transferability to the labourer and tightens the labourer's participation constraint. Bank j chooses the fee by balancing these effects.

As in the autarkic contract, it is always optimal for bank j to set $\chi_{j,j} = 0$. The only constraint that $\chi_{j,j}$ enters is the labourer's participation constraint. Raising the intrabank fee tightens that constraint more than it increases fee revenue, so it is always more efficient to hold the fee to zero and expand investment instead.

Bank j can choose its own unit of account so that the initial promise-keeping constraint, equation (9), is satisfied. One such choice is the natural unit of account, $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$, which gives $\epsilon_{j,j} = 1$. The own-bank promise-keeping constraint is then slack under the optimal transferable-debt contract unless the optimal fee equals $1 - 1/\epsilon_{j,k}$, in which case

equations (9) and (10) both bind. More generally, the feasible range of bank j's unit-of-account choices depends on the equilibrium contract, which in turn depends on bank k's choice.

The fee cutoffs derived in Appendix C imply three regimes for the transferable-debt contract. Bank j charges no fee when $\epsilon_{j,k} \leq \bar{\epsilon}^{(0)}$, an interior fee when $\bar{\epsilon}^{(0)} < \epsilon_{j,k} < \bar{\epsilon}^{(1)}$, and the maximum feasible fee $\gamma/(1+\gamma)$ when $\epsilon_{j,k} \geq \bar{\epsilon}^{(1)}$, provided that bank j continues to allow interbank transfers. Once the interbank constraint binds, bank profit is strictly decreasing in $\epsilon_{j,k}$ (Appendix B).

In the intermediate case $\bar{\epsilon}^{(0)} < \epsilon_{j,k} < \bar{\epsilon}^{(1)}$, the optimal transfer fee $\chi^*_{j,k}$ is given in closed form in Appendix C. Section 4 specialises this result to the symmetric two-state model, in which the settlement requirement can be expressed as a function of ρ.

3.3 Interoperability Choice

The next step is to determine whether bank j chooses to allow its liabilities to circulate beyond its own region. This decision compares the profit from transferable debt with the profit from autarky. Bank j's decision can be formalised as

$$\pi_j\left(\epsilon_{j,k}\right) = \max\left\{W_j\left(\epsilon_{j,k}\right), V_j^{(0)}\right\}, \tag{26}$$

where $W_j\left(\epsilon_{j,k}\right)$ denotes bank j's profit if it allows its liabilities to be transferred to region k with

$$W_j\left(\epsilon_{j,k}\right) = \begin{cases} \varphi^{\frac{\varphi}{1-\varphi}}\left(1+\gamma(\alpha+\beta)\right)^{\frac{\varphi}{1-\varphi}}(1-\varphi) & \text{if } \epsilon_{j,k} \leq \frac{1}{\varphi} \\ \left[\frac{1+\gamma\alpha+\beta\left(\gamma-(1+\gamma)\chi^*_{j,k}(\epsilon_{j,k})\right)}{\epsilon_{j,k}\left(1-\chi^*_{j,k}(\epsilon_{j,k})\right)}\right]^{\frac{\varphi}{1-\varphi}}\left(1-\frac{1-\beta\chi^*_{j,k}(\epsilon_{j,k})}{\epsilon_{j,k}\left(1-\chi^*_{j,k}(\epsilon_{j,k})\right)}\right) & \text{if } \epsilon_{j,k} > \frac{1}{\varphi} \end{cases} \tag{27}$$

and where $V_j^{(0)}$ is bank j's profit if it chooses autarky as defined in Proposition 1.

Bank k's choice of unit of account affects the settlement requirement $\epsilon_{j,k}$ faced by bank j. Bank j responds by choosing the terms of its debt contract and whether to allow its debt to be transferred to region k.

When $\epsilon_{j,k} \leq 1/\varphi$, bank j strictly prefers to allow interbank transfers because its profit exceeds the autarkic payoff. As $\epsilon_{j,k}$ increases beyond this point, the profit from transferable debt declines. In the limit,

$$\lim_{\epsilon_{j,k}\to\infty} W_j\left(\epsilon_{j,k}\right) = 0,$$

so bank j eventually earns a higher profit under autarky.

The following proposition formalises this trade-off by establishing a unique threshold above which autarky dominates.

Proposition 3 (Autarky Threshold)**.** *There exists a unique threshold* $\bar{\epsilon}^{(2)}_{j,k} > \frac{1}{\varphi}$*, implicitly defined by* $W_j\left(\bar{\epsilon}^{(2)}_{j,k}\right) = V^{(0)}_j$*, such that:*

(i) bank j *strictly prefers allowing interbank transfers when* $\epsilon_{j,k} < \bar{\epsilon}^{(2)}_{j,k}$*;*

(ii) bank j *strictly prefers autarky when* $\epsilon_{j,k} > \bar{\epsilon}^{(2)}_{j,k}$*.*

The threshold satisfies

$$\bar{\epsilon}^{(2)}_{j,k} = \bar{\epsilon}^{(2)}(\gamma, \alpha, \beta, \varphi)$$

and is strictly increasing in β *and in* γ*.*

Proof. See Appendix B. □

At $\epsilon_{j,k} = \bar{\epsilon}^{(2)}_{j,k}$, bank j is indifferent between allowing interbank transfers and choosing autarky. The threshold therefore identifies the point beyond which the gains from cross-region trade no longer compensate the bank for the associated settlement burden. A higher probability of meeting a foreign artisan or a larger gain from trade raises the value of allowing interbank transfers, so the threshold increases with both β and γ.

Figure 1 illustrates how bank j's best response varies with the settlement requirement induced by bank k's unit-of-account choice. Bank j chooses its own unit of account so that the initial promise-keeping constraint in equation (9) is satisfied. The figure therefore isolates how bank j's willingness to permit interbank transfers changes with $\epsilon_{j,k}$.

Figure 1 illustrates the conditions under which bank j restricts transferability as the settlement requirement $\epsilon_{j,k}$ increases.

Consider bank k as an incumbent with an established unit of account and bank j as an entrant that takes this choice as given. When the resulting settlement requirement satisfies $\epsilon_{j,k} \leq 1/\varphi$, the entrant can implement the constrained-optimal contract and sets $\chi_{j,k} = 0$. The interbank promise-keeping constraint is slack when $\epsilon_{j,k} < 1/\varphi$ and holds with equality at $\epsilon_{j,k} = 1/\varphi$. As $\epsilon_{j,k}$ rises above this threshold, the obligation denominated in bank k's unit of account becomes more costly relative to the value of bank j's production in its worst-case price realisation. The entrant first reduces investment and, once $\epsilon_{j,k}$ exceeds $\bar{\epsilon}^{(0)}_{j,k}$, raises the interbank transfer fee. When $\epsilon_{j,k}$ exceeds $\bar{\epsilon}^{(2)}_{j,k}$, the entrant no longer permits interbank transfers and chooses autarky. This restriction on transferability is an endogenous response to the decline in the profitability of transferable debt.

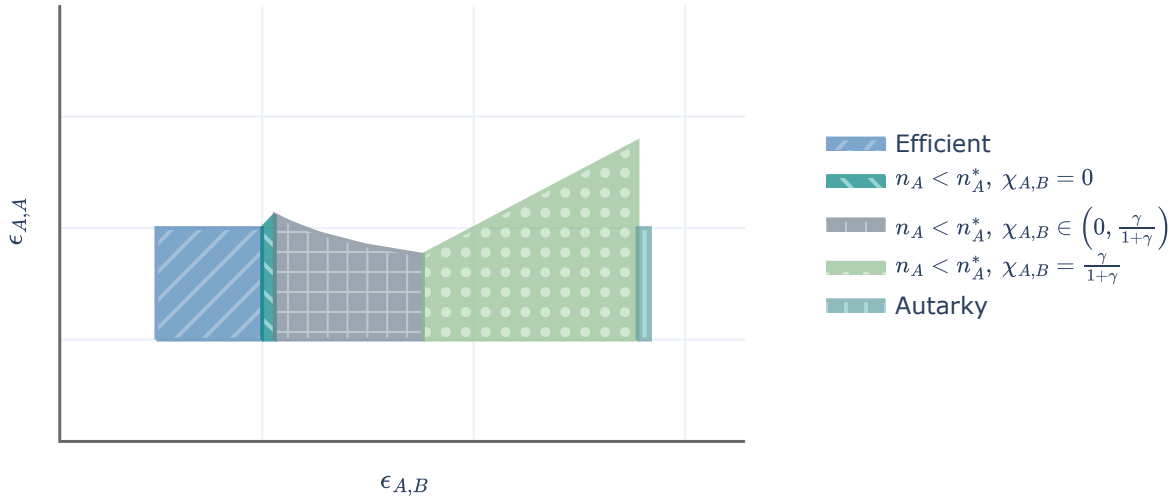


Figure 1: Bank j's best responses to the settlement requirement induced by bank k's choice of unit of account. The horizontal axis shows the cross-bank settlement requirement $\epsilon_{A,B}$ and the vertical axis shows the own-bank settlement requirement $\epsilon_{A,A}$.

3.4 Strategic Interaction of Banks

I now characterise the strategic interaction between banks over their choice of unit of account. I consider two equilibrium concepts. The first is a staggered-entry benchmark in which bank k has committed to a unit of account before bank j enters and bank j best-responds to bank k's choice. The second is a simultaneous-move Nash equilibrium in which both banks choose their units of account, transfer fees, and whether to allow interbank transfers. In both cases, each bank retains its choice over transfer fees and transferability.

Consider first the staggered-entry case, in which bank k has already committed to its unit of account before bank j enters. Bank j then takes $\hat{\mathbf{b}}_k$ as given and chooses its contract as in Propositions 1–3. The relevant object is the settlement requirement

$$\epsilon_{j,k} = \max_{\mathbf{p} \in \mathbf{P}} \frac{\mathbf{p}' \hat{\mathbf{b}}_k}{\mathbf{p}' \hat{\mathbf{y}}_j}.$$

If $\epsilon_{j,k} \leq 1/\varphi$, the entrant implements the constrained optimum by setting $\chi_{j,k} = 0$.

I now turn to the Nash equilibrium. Efficiency requires all four settlement requirements, $\epsilon_{j,j}$, $\epsilon_{j,k}$, $\epsilon_{k,j}$ and $\epsilon_{k,k}$, to be at most $1/\varphi$. When feasible unit-of-account choices satisfy these inequalities for both banks, an efficient fee-free equilibrium exists. The following proposition records the optimal contract regimes induced by a given pair $(\epsilon_{j,k}, \epsilon_{k,j})$ and the corresponding equilibrium existence results.

Proposition 4 (Nash Equilibrium)**.** *Consider the simultaneous-move game in which each bank $i \in \{j, k\}$ chooses $\hat{\mathbf{b}}_i$ and optimises its debt contract as in Propositions 1–3, yielding payoff $\pi_i(\epsilon_{i,\ell}) = \max\{W_i(\epsilon_{i,\ell}), V_i^{(0)}\}$. Both banks choosing autarky is a Nash equilibrium if and only if, for each bank i, $\epsilon_{i,i} \leq 1/\varphi$ and $\epsilon_{i,\ell} \geq \bar{\epsilon}_{i,\ell}^{(2)}$. Given the other bank's unit of account, bank i's optimal contract depends on $\epsilon_{i,\ell}$ and falls into one of four regimes. For a given $\epsilon_{i,\ell}$, the optimal fee $\chi_{i,\ell}^*$ of a transferable contract is uniquely determined.*

- Zero-fee: $\epsilon_{i,\ell} \leq \bar{\epsilon}_{i,\ell}^{(0)}$, $\chi_{i,\ell}^* = 0$. *Investment is at the constrained optimum if $\epsilon_{i,\ell} \leq 1/\varphi$; the interbank constraint binds and investment falls below the optimum for $\epsilon_{i,\ell} \in (1/\varphi, \bar{\epsilon}_{i,\ell}^{(0)}]$.*
- Positive-fee: $\bar{\epsilon}_{i,\ell}^{(0)} < \epsilon_{i,\ell} < \bar{\epsilon}_{i,\ell}^{(1)}$, $\chi_{i,\ell}^* \in (0, \gamma/(1+\gamma))$.
- Maximum-fee: $\bar{\epsilon}_{i,\ell}^{(1)} \leq \epsilon_{i,\ell} < \bar{\epsilon}_{i,\ell}^{(2)}$, $\chi_{i,\ell}^* = \gamma/(1+\gamma)$.
- Autarky: $\epsilon_{i,\ell} \geq \bar{\epsilon}_{i,\ell}^{(2)}$, *interbank transfers prohibited; at equality, the maximum-fee contract is equally optimal.*

If both banks choose a common unit of account $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k = \hat{\mathbf{b}}$ such that $\max_{\mathbf{p}\in\mathbf{P}} \mathbf{p}'\hat{\mathbf{b}}/\mathbf{p}'\hat{\mathbf{y}}_i \leq 1/\varphi$ for both banks, the resulting profile is an efficient fee-free Nash equilibrium.

Proof. See Appendix B. □

Efficient fee-free equilibria, when they exist, Pareto-dominate the other contract outcomes described above. A common unit of account supports an efficient equilibrium when the shared basket satisfies the settlement bound in Proposition 4 for both banks, but a common unit is not necessary for efficiency. Distinct units of account can also implement the constrained optimum if each bank's own-bank and cross-bank settlement requirements are at most $1/\varphi$. Zero fees are not sufficient for efficiency. When $\epsilon_{i,\ell} \in (1/\varphi, \bar{\epsilon}_{i,\ell}^{(0)}]$, the interbank promise-keeping constraint binds and investment falls below the constrained optimum even though no fee is charged.

Equilibrium multiplicity can arise in the baseline environment because multiple unit-of-account choices can satisfy the relevant promise-keeping constraints while allowing the same optimal contract, debt capacity, and profit. Banks are therefore indifferent across a range of feasible unit-of-account choices. Section 6 considers an extension in which acceptance depends on the worst-state value of debt, which reduces this indifference.

3.5 Common Unit of Account and Deviations from Par

A liability denominated in unit of account $\hat{\mathbf{b}}_j$ is a state-contingent claim with per-unit value $\mathbf{p}'\hat{\mathbf{b}}_j$ in state $\mathbf{p}$. When both banks adopt a common unit of account, $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k$, their liabilities have the same per-unit value in every state. This establishes parity but does not by itself ensure efficiency. The common unit removes the additional settlement friction arising from differences between their effective denominations. For bank j, the interbank settlement requirement then equals its own settlement requirement, so $\epsilon_{j,k} = \epsilon_{j,j}$. If the shared basket is more valuable than bank j's unit of production for some price realisation, then $\epsilon_{j,j} > 1$ and the promise-keeping constraint may still bind. A common unit therefore removes the additional cross-denomination exposure without necessarily eliminating the settlement burden associated with the shared repayment basket.

Under the stablecoin interpretation, two liabilities can share a nominal dollar denomination while retaining different effective denominations. If the banks choose a common effective denomination, the liabilities have identical per-unit values in every state and are therefore valued at par in the model's no-default environment. Different effective denominations can produce different state-contingent values, although whether they do so depends on the price support. The date-1 conversion rate remains based on expected claim values and is distinct from this realised state-contingent comparison.

Parity and efficiency are not equivalent. Efficiency requires each bank's own-bank and cross-bank settlement requirements to be at most $1/\varphi$, allowing investment to reach the constrained-optimal level. These conditions can hold when banks use different units of account, so parity is not necessary for efficiency. Conversely, a common unit of account supports efficient investment only when the shared basket satisfies the settlement bound for both banks. The liabilities may be valued at par while the banks' own promise-keeping constraints bind and investment remains below the constrained optimum. Parity is therefore neither necessary nor sufficient for efficiency.

When the settlement requirements generated by feasible unit-of-account choices are sufficiently large, banks may respond by charging positive transfer fees or choosing autarky. The resulting reductions in investment or trade lower welfare relative to the constrained optimum. Section 5 studies policy interventions that alter these settlement conditions.

4. Two-State Analytical Example

This section studies a two-state, two-good version of the model in which all equilibrium objects can be expressed as functions of a single parameter ρ governing relative-price

dispersion. The two-state case makes the equilibrium structure transparent and can be interpreted as two issuers of USD stablecoins. Their liabilities share a nominal dollar denomination, but each bank chooses the basket promised in repayment and therefore the effective denomination of its liability. The analysis compares the natural-unit-of-account benchmark with equilibria that use a common unit of account, without restricting banks to these two denomination choices.

4.1 Setup

To fix ideas, I assume that bank j produces only good A and bank k produces only good B. For convenience, I index each bank by the good it produces, replacing the subscripts j and k with A and B. Their units of production are

$$\hat{\mathbf{y}}_A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{y}}_B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{28}$$

I restrict the price process to two states that occur with equal probability. In state A, $p_A = 1 - \rho_A$ and $p_B = 1 + \rho_B$; in state B, $p_A = 1 + \rho_A$ and $p_B = 1 - \rho_B$. The parameters $\rho_A, \rho_B \in [0, 1)$ determine the dispersion of each good's price across the two states. The two price vectors span the convex price set $\mathbf{P}$. Imposing promise keeping at both vectors is equivalent to imposing it throughout $\mathbf{P}$.

In the symmetric case, $\rho_A = \rho_B = \rho$. Under the natural unit of account, each bank denominates its debt in its own unit of production, so that $\hat{\mathbf{b}}_A = \hat{\mathbf{y}}_A$ and $\hat{\mathbf{b}}_B = \hat{\mathbf{y}}_B$. Specialising the settlement requirement $\epsilon_{j,k}$ from equation (8) to this denomination choice and the two-state price process yields

$$\epsilon_{A,B} \;=\; \epsilon_{B,A} \;=\; \frac{1+\rho}{1-\rho}, \qquad \rho \in [0, 1). \tag{29}$$

The mapping in equation (29) is strictly increasing in ρ. It therefore allows the fee and autarky cutoffs derived in ϵ-space to be translated directly into cutoffs in ρ-space.

4.2 Nash Equilibrium Characterisation

Under the natural-unit-of-account benchmark, where $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$, the mapping in equation (29) reduces the general Nash equilibrium characterisation in Proposition 4 and the

fee cutoffs derived in Appendix C to functions of ρ alone. Define

$$\Gamma \equiv \frac{\gamma(1-\varphi)\beta}{1+\gamma\alpha-\beta}. \tag{30}$$

Let $\rho^{(0)}$, $\rho^{(1)}$, and $\rho^{(2)}$ denote the values of ρ obtained by mapping the corresponding cutoffs $\bar{\epsilon}^{(0)}$, $\bar{\epsilon}^{(1)}$, and $\bar{\epsilon}^{(2)}$ through equation (29). Thus

$$\rho^{(0)} = \frac{\bar{\epsilon}^{(0)}-1}{\bar{\epsilon}^{(0)}+1}, \qquad \bar{\epsilon}^{(0)} = \frac{1}{\varphi}\big(1+\Gamma(1-\alpha-\beta)\big), \tag{31}$$

$$\rho^{(1)} = \frac{\bar{\epsilon}^{(1)}-1}{\bar{\epsilon}^{(1)}+1}, \tag{32}$$

$$\rho^{(2)} = \frac{\bar{\epsilon}^{(2)}-1}{\bar{\epsilon}^{(2)}+1}. \tag{33}$$

Proposition 3 defines $\bar{\epsilon}^{(2)}$ implicitly through $W_j(\bar{\epsilon}^{(2)}) = V_j^{(0)}$, so $\rho^{(2)}$ has no closed form. Under the model's parameter restrictions,

$$0 < \rho^{(0)} < \rho^{(1)} < \rho^{(2)} < 1.$$

Finally, let $\rho_0 \equiv (1-\varphi)/(1+\varphi)$ denote the value at which $\epsilon(\rho) = 1/\varphi$. The symbol ρ_0 refers to the promise-keeping threshold. By contrast, $\rho^{(0)}$ denotes the zero-fee cutoff. Since the zero-fee region extends beyond the point at which the interbank constraint first binds, $\rho_0 < \rho^{(0)}$. The following proposition characterises equilibrium outcomes under the natural unit of account.

Proposition 5 (Two-State Natural Unit of Account)**.** *In the symmetric two-state model under the natural unit of account, with $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$, let $\rho^{(0)}$, $\rho^{(1)}$, and $\rho^{(2)}$ be defined by equations* (31)–(33)*.*

(a) *($\rho \le \rho_0$): Both banks set $\chi^* = 0$, investment is at the constrained optimum, and all equilibria are efficient regardless of the unit of account.*

(b) *($\rho_0 < \rho \le \rho^{(0)}$): Both banks set $\chi^* = 0$, but investment is below the constrained optimum and continuously declining in ρ.*

(c) *($\rho^{(0)} < \rho \le \rho^{(2)}$): Both banks charge $\chi^*(\rho) > 0$. For $\rho^{(0)} < \rho < \rho^{(1)}$, the fee is interior, $\chi^*(\rho) \in (0, \gamma/(1+\gamma))$, strictly increasing in ρ. For $\rho^{(1)} \le \rho \le \rho^{(2)}$, the fee is at its maximum, $\chi^* = \gamma/(1+\gamma)$. At $\rho = \rho^{(2)}$, autarky is equally optimal.*

(d) *($\rho > \rho^{(2)}$): Both banks choose autarky; no equilibrium with interbank transfers exists under the natural unit of account.*

Proof. See Appendix D. □

As ρ increases through cases (b)–(d), the settlement requirement rises and banks respond first with lower investment, then with transfer fees, and finally with autarky. Under the natural unit of account, low values of ρ generate limited departures from par that can be accommodated without constraining investment. When ρ exceeds ρ_0, investment falls below the constrained optimum even though transfers initially remain fee-free. A further increase in ρ makes positive transfer fees optimal, before sufficiently large settlement requirements lead both banks to choose autarky. Case (a) establishes the stronger result that, when $\rho \leq \rho_0$, the natural unit of account maximises the interbank settlement requirement over all feasible unit-of-account choices, so all equilibria are efficient regardless of the unit of account.

4.3 Common Unit of Account Equilibria

The natural-unit-of-account benchmark fixes each bank's unit of account at $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$. A common unit of account instead requires the banks to choose the same repayment basket, so that $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k = [\omega, 1-\omega]'$. In that case, the interbank settlement requirement $\epsilon_{j,k}$ is equal to $\epsilon_{j,j}$, the settlement requirement associated with bank j's own unit of account relative to its production. By symmetry of the two banks' units of production, $\omega = 1/2$ minimises $\max\{\epsilon_{A,A}(\omega), \epsilon_{B,B}(\omega)\}$. I use the equal-weight basket to assess whether a common unit can satisfy the relevant settlement bounds for both banks. At this basket, both banks face $\epsilon_{j,j} = 1/(1-\rho)$, which is strictly less than $(1+\rho)/(1-\rho)$ for all $\rho > 0$. Define

$$\rho_c \;\equiv\; 1 - \frac{1}{\bar{\epsilon}^{(0)}} \;=\; \frac{\bar{\epsilon}^{(0)} - 1}{\bar{\epsilon}^{(0)}}, \tag{34}$$

which is the value at which the equal-weight basket's settlement requirement, $1/(1-\rho)$, equals the zero-fee cutoff $\bar{\epsilon}^{(0)}$. The ordering $\rho^{(0)} < \rho_c$ follows from $\bar{\epsilon}^{(0)} < \bar{\epsilon}^{(0)} + 1$.

Proposition 6 (Common Unit of Account Equilibria)**.** *In the symmetric two-state model:*

(a) *($\rho \leq 1 - \varphi$): There exist Nash equilibria in which both banks adopt a common unit of account. An efficient such equilibrium exists.*

(b) *($1 - \varphi < \rho \leq \rho_c$): There exist Nash equilibria in which both banks adopt a common unit of account, but all such equilibria are inefficient.*

(c) *($\rho > \rho_c$): No Nash equilibrium with a common unit of account exists.*

Proof. See Appendix D. □

Proposition 6 separates the existence of a common-unit equilibrium from its efficiency. For $\rho \leq 1 - \varphi$, the equal-weight basket supports an efficient common-unit equilibrium. For $1 - \varphi < \rho \leq \rho_c$, common-unit equilibria continue to exist, but none implements the constrained optimum. Even the equal-weight basket, which minimises the larger of the two banks' settlement requirements, then gives $\epsilon_{j,j} = 1/(1-\rho) > 1/\varphi$ for both banks, so investment is constrained. The common effective denomination nevertheless gives the two liabilities the same state-contingent value. When the banks are interpreted as issuers of USD stablecoins, their liabilities are therefore valued at par despite the investment distortion. This demonstrates that parity is not sufficient for efficient exchange. For $\rho > \rho_c$, a unilateral deviation to the natural unit of account is strictly profitable, so no common-unit equilibrium exists.

4.4 Welfare Analysis

Welfare comparisons in this subsection are conducted under the natural-unit-of-account benchmark, $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$. In the symmetric two-state model, this benchmark yields the largest cross-bank settlement friction among feasible unit-of-account choices. It therefore gives the lowest-welfare Nash equilibrium with interbank transfers, so welfare losses under any other equilibrium are weakly smaller. This ordering follows from symmetry and Proposition 4. The welfare loss is measured relative to the constrained-efficient benchmark W^* defined in Section 2. An efficient equilibrium attains this benchmark when $\chi^* = 0$ and investment reaches the constrained-optimal level n^*. Since labourers and artisans receive no surplus in these equilibria, per-bank welfare equals bank profit $W(\rho)$, and the welfare loss is $\Delta W(\rho) \equiv W^* - W(\rho)$.

Proposition 7 (Welfare Loss under the Natural Unit of Account)**.** *In the symmetric two-state model under the natural unit of account ($\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$), the per-bank welfare loss $\Delta W(\rho) \equiv W^* - W(\rho)$ relative to the constrained-efficient benchmark satisfies:*

(a) $\Delta W(\rho) = 0$ *for* $\rho \leq \rho_0$ *(efficient equilibria).*

(b) $\Delta W(\rho) > 0$ *and strictly increasing for* $\rho_0 < \rho \leq \rho^{(2)}$ *(zero-fee suboptimal and fee-charging cases).*

(c) $\Delta W(\rho) = W^* - V_j^{(0)}$ *(constant) for* $\rho > \rho^{(2)}$ *(autarky).*

Moreover, $\Delta W(\rho)$ is continuous on $[0, \rho^{(2)}]$ and constant for $\rho > \rho^{(2)}$. Explicit welfare expressions are provided in Appendix D.

Proof. See Appendix D. □

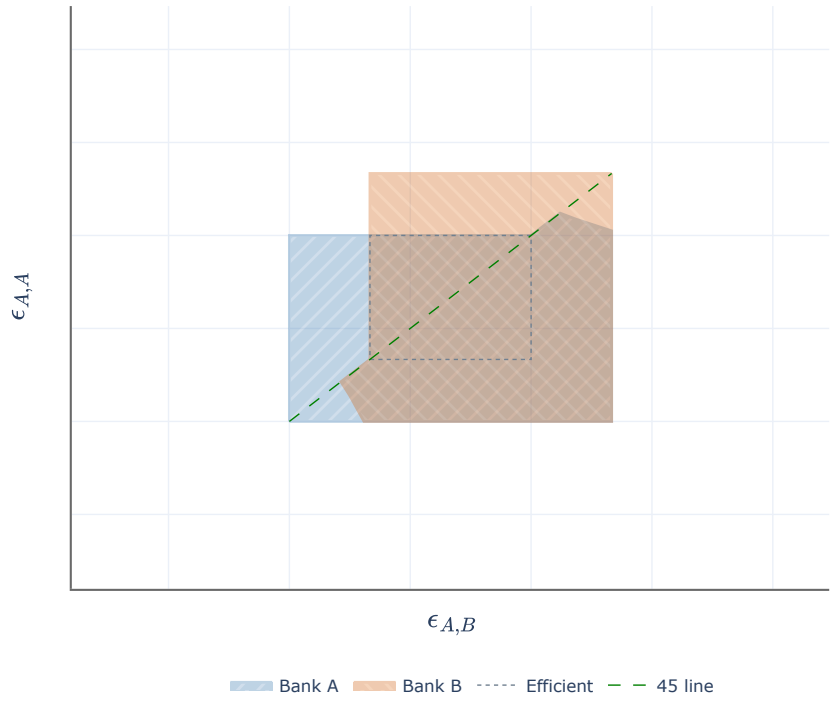


Figure 2: Nash equilibrium sets for $\rho_A = \rho_B = 0.4$, with $\varphi = 0.5$, $\gamma = 1.0$, $\alpha = 0.3$, and $\beta = 0.3$. The horizontal axis shows $\epsilon_{A,B}$ and the vertical axis shows $\epsilon_{A,A}$. The grey dotted boundary marks the efficient subset, while the green dashed 45-degree line marks common-unit-of-account choices.

The proposition shows how welfare under the natural-unit-of-account benchmark declines as relative-price dispersion rises. At low values of ρ, the equilibrium attains the constrained-efficient benchmark. Once ρ exceeds ρ_0, investment becomes inefficient even though transfers initially remain fee-free. Further increases in ρ lead to positive transfer fees and eventually to autarky.

4.5 Illustrations

Figures 2 and 3 illustrate the equilibrium sets in the two-state case for two values of ρ. The natural unit of account and the common unit of account are useful benchmark cases within these figures, but the equilibrium set also includes intermediate unit-of-account choices.

The figures use the parameter values $\varphi = 0.5$, $\gamma = 1.0$, $\alpha = 0.3$, and $\beta = 0.3$. These values illustrate the transition from efficient exchange to positive transfer fees and, at sufficiently high relative-price dispersion, to autarky.

Figure 2 illustrates the case $\rho = 0.4$. By Proposition 5(c), this value lies in the positive-

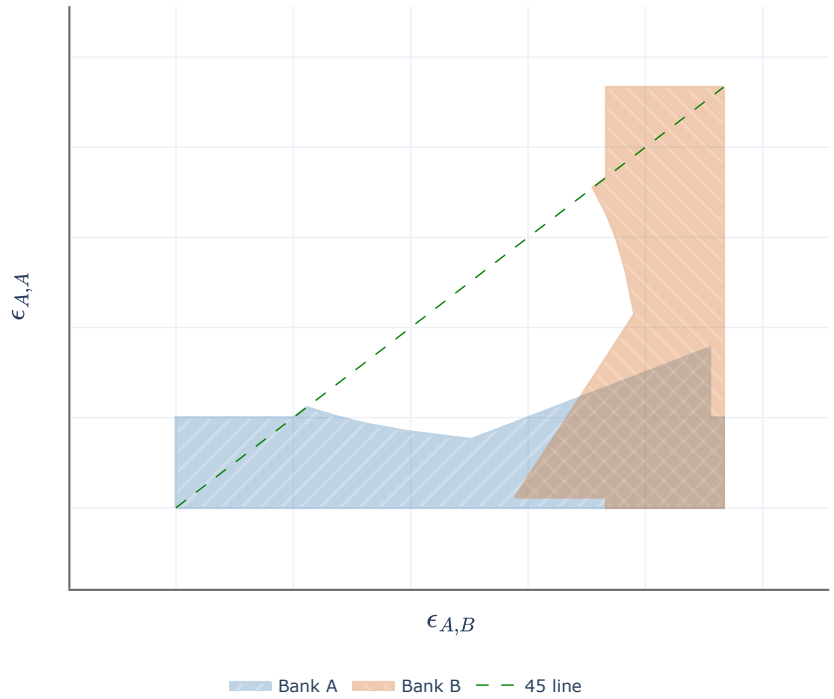


Figure 3: Nash equilibrium sets for $\rho_A = \rho_B = 0.7$, with $\varphi = 0.5$, $\gamma = 1.0$, $\alpha = 0.3$, and $\beta = 0.3$. The horizontal axis shows $\epsilon_{A,B}$ and the vertical axis shows $\epsilon_{A,A}$. The green dashed 45-degree line is the common-unit-of-account reference locus.

fee case under the natural unit of account, so both banks charge positive transfer fees and investment is below the constrained optimum. The grey dotted boundary marks the efficient subset. Proposition 6(a) shows that efficient equilibria still exist when banks coordinate on a common unit of account, shown by the green dashed 45-degree line. A common unit of account removes the additional settlement wedge created by differences between the banks' effective denominations, so interbank transfer fees are zero in these equilibria.

Figure 3 illustrates the case $\rho = 0.7$. Proposition 5(d) implies that, under the natural unit of account, both banks choose autarky because $\rho > \rho^{(2)}$. Proposition 6(c) shows that no common-unit-of-account equilibrium exists at this value of ρ.

5. Policy Analysis

This section studies how policy can affect equilibrium outcomes. It first considers a mandate requiring issuers to adopt a common unit of account. It then examines two forms of public settlement: central bank reserves and cash.

The model's objects have natural policy interpretations. Differences in banks' production baskets $\hat{\mathbf{y}}_j$ capture heterogeneity in the resources available to intermediaries. The settlement requirement $\epsilon_{j,k}$ measures the cost to bank j of honouring claims denominated in bank k's unit of account in its worst-case price realisation, while $\chi_{j,k}$ is the contractual fee charged on an interbank transfer. The reserve parameter τ determines the share of production risk pooled through the monetary unit $\hat{\mathbf{y}}_m$. Cash denominated in $\hat{\mathbf{y}}_m$ provides an alternative settlement channel when direct interbank transfer is costly.

The analysis focuses on singleness within a monetary system. Global singleness raises additional questions concerning monetary sovereignty and exchange-rate policy, which lie outside the model. When banks are interpreted as issuers of private digital money, differences across offshore or foreign-currency issuers can be represented through their effective denominations and the composition of their resources. These considerations differ from the within-system question studied here, where par valuation and transferability across issuers are policy objectives.

The policy instruments operate through two margins. A common-unit mandate restricts issuers' choices of effective denomination. Public settlement arrangements instead alter the conditions under which cross-region obligations are honoured. Reserves pool part of the state-contingent variation in issuer resources, whereas cash provides an alternative settlement channel.

5.1 Mandating a Common Unit of Account

A common-unit-of-account mandate requires banks to use the same effective denomination. When the banks are interpreted as issuers of USD stablecoins, this requirement is stronger than assigning both liabilities a nominal US dollar denomination. When both banks set $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_a$, their liabilities have the same per-unit state-contingent value. The additional settlement exposure arising from different units of account therefore disappears, and interbank transfer fees are zero. The mandated unit $\hat{\mathbf{y}}_a$ need not coincide with the monetary unit $\hat{\mathbf{y}}_m$. The former is a regulatory choice of effective denomination, whereas the latter is the unit in which reserves and cash are denominated. If $\hat{\mathbf{y}}_a$ generates a large own-bank settlement requirement relative to bank j's production basket $\hat{\mathbf{y}}_j$, the mandate may tighten the bank's own promise-keeping constraint.

A common-unit-of-account mandate therefore requires banks to set

$$\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_a. \tag{35}$$

Substituting this restriction into the promise-keeping constraints (9) and (10) shows that there is no gain from setting $\chi_{j,k} > 0$, and bank k faces the symmetric problem. If the mandated basket $\hat{\mathbf{y}}_a$ corresponds to a common-unit-of-account Nash equilibrium, the mandate implements that equilibrium. Otherwise, it imposes a denomination profile that private choices would not sustain.

The mandate nevertheless creates two policy risks. The first is that the choice of the common unit of account may matter. Figure 2 illustrates a case in which an efficient common-unit-of-account equilibrium exists. A mandate that selects a different common basket can still impose an inefficient outcome.

The second risk arises when no common-unit-of-account Nash equilibrium exists without intervention. Figure 3 illustrates one such case. A mandate can still enforce a common denomination by preventing unilateral departures from the mandated basket, but it does not remove the banks' own promise-keeping constraints. If the mandated unit generates an own-bank settlement requirement above the efficient bound, banks must reduce borrowing below the optimum. Appendix E provides the formal characterisation.

Proposition 6 characterises the private common-unit equilibria. For $\rho \leq \rho_c$, a common-unit-of-account Nash equilibrium exists, so a mandate can serve as an equilibrium-selection device when it imposes a basket that constitutes such an equilibrium. For $\rho \leq 1-\varphi$, the symmetric basket at $\omega = 1/2$ supports efficient investment. For $\rho > \rho_c$, no common-unit-of-account Nash equilibrium exists, although a mandate can still enforce a common basket and may constrain investment. Appendix E shows that, in the symmetric two-state model, the mandated weight $\omega = 1/2$ minimises the larger of the two banks' settlement requirements.

The analysis distinguishes mandates that select a privately sustainable common-basket equilibrium from those that impose a basket that private choices would not sustain. The competitive process through which a common unit might emerge lies outside the model. When $\rho \leq 1 - \varphi$, a mandate imposing the symmetric basket can attain the constrained-efficient benchmark. When $\rho > \rho_c$, a mandate can enforce a common unit, but it cannot eliminate the own-bank settlement burden associated with that basket.

5.2 Central Bank Reserves

Central bank reserves are represented by a tax-and-transfer system that pools a fraction τ of each bank's production risk through the monetary unit $\hat{\mathbf{y}}_m$. The tax-and-transfer construction provides a stylised representation of settlement in central bank money. Formally, the government taxes a fraction $\tau \in [0, 1]$ of each bank's output. The tax rate

is announced at date 0 and collected at date 2. The government does not consume the proceeds. Instead, it returns claims of equal expected value to the banks through the common monetary basket. Implementation is enforced by issuing each bank at date 0 a tax obligation and a corresponding transfer claim payable at date 2. I assume that the government can honour these state-contingent transfer claims using its broader fiscal capacity, so the scheme is required to balance in expectation rather than state by state. This fiscal backing does not enter banks' resources or payoffs. The construction abstracts from collateralised central bank liquidity facilities and interest paid on reserves.

This intervention is ex ante neutral because each bank receives a transfer equal in expected value to its tax payment. The tax and transfer claims nevertheless have different units of account. Bank j pays taxes in its production basket $\hat{\mathbf{y}}_j$, while the transfer is denominated in the monetary basket $\hat{\mathbf{y}}_m$, where

$$\hat{\mathbf{y}}_m = \frac{1}{2}\left(\hat{\mathbf{y}}_j + \hat{\mathbf{y}}_k\right). \tag{36}$$

The equal-weight monetary basket is the symmetric midpoint of the two production baskets. In the symmetric two-state model, its value is invariant across states. The tax-and-transfer system therefore provides partial insurance against relative-price shocks by pooling part of each bank's production risk through a common settlement asset.

The policy changes the set of competitive equilibria through its effect on bank resources. At the beginning of date 2, bank j has its production output net of taxes together with the transfer claim. Its effective resource basket is therefore

$$\tilde{\mathbf{y}}_j = (1-\tau)\,\hat{\mathbf{y}}_j + \tau\hat{\mathbf{y}}_m. \tag{37}$$

For the post-policy natural-unit benchmark, each bank denominates its liabilities in its effective resource basket, so $\hat{\mathbf{b}}_j = \tilde{\mathbf{y}}_j$. This is the analogue of the natural-unit choice after the reserve intervention. It does not change the pre-policy definition used in Section 4.

Under this benchmark, central bank reserves reduce the effective relative-price dispersion from ρ to $\tilde{\rho}(\tau) = \rho(1-\tau)$. This shifts the cutoffs in Propositions 5 and 6, expanding both the efficient-equilibrium existence region and the post-policy natural-unit zero-fee region. The resulting comparative statics are summarised in Proposition 8.

Proposition 8 (Central Bank Reserves)**.** *In the symmetric two-state model with reserve requirement $\tau \in [0,1]$, the effective price dispersion parameter is $\tilde{\rho}(\tau) = \rho(1-\tau)$ and the state-contingent settlement cost becomes*

$$\epsilon(\rho,\tau) \;=\; \frac{1+\rho(1-\tau)}{1-\rho(1-\tau)}. \tag{38}$$

Each reserve threshold below, and τ_c^* in equation (42), is zero at $\rho = 0$.

(a) *An efficient equilibrium exists if and only if $\tilde{\rho}(\tau) \leq 1 - \varphi$, equivalently $\tau \geq \tau^{eff}$ where*

$$\tau^{eff} \;=\; \max\left\{0,\; 1 - \frac{1-\varphi}{\rho}\right\}. \tag{39}$$

Under the natural unit of account, zero fees obtain if and only if $\tilde{\rho}(\tau) \leq \rho^{(0)}$, equivalently $\tau \geq \tau^*$ where

$$\tau^* \;=\; \max\left\{0,\; 1 - \frac{\rho^{(0)}}{\rho}\right\}. \tag{40}$$

(b) *Under the natural unit of account, transfers are optimal if and only if $\tilde{\rho}(\tau) \leq \rho^{(2)}$, equivalently $\tau \geq \tau^A \equiv \max\{0,\, 1-\rho^{(2)}/\rho\}$; at $\tilde{\rho}(\tau) = \rho^{(2)}$, autarky is equally optimal.*

Proof. Substituting $\rho(1-\tau)$ for ρ in (29) gives $\epsilon(\rho,\tau) = (1+\rho(1-\tau))/(1-\rho(1-\tau))$. Under the equal-weight common basket $\omega = 1/2$, both banks face $\epsilon_{j,j} = 1/(1-\tilde{\rho})$. By Proposition 4, efficient equilibria exist if and only if $1/(1-\tilde{\rho}) \leq 1/\varphi$, i.e. $\tilde{\rho}(\tau) \leq 1-\varphi$, yielding $\tau \geq 1-(1-\varphi)/\rho \equiv \tau^{eff}$. Under the natural unit of account, $\epsilon \leq \bar{\epsilon}^{(0)}$ yields $\tau \geq 1-\rho^{(0)}/\rho \equiv \tau^*$. Part (b) follows from Proposition 3, since $\epsilon(\rho,\tau) \leq \bar{\epsilon}^{(2)}$ if and only if $\tilde{\rho}(\tau) \leq \rho^{(2)}$. □

Setting $\tau = 1$ ensures that an efficient equilibrium exists for every $\rho \in [0,1)$. This limiting case illustrates the settlement logic of narrow banking under the maintained assumption that reserve provision is costless. The analysis does not include the costs of a 100% reserve requirement. When all production risk is mutualised through the common monetary basket, the two banks have the same effective resource basket and the lowest attainable settlement requirement in this policy construction.

5.3 Cash

Cash provides an alternative public settlement claim for cross-region transactions. In the symmetric two-state model, cash is denominated in the monetary basket

$$\hat{\mathbf{y}}_m = \frac{1}{2}\left(\hat{\mathbf{y}}_A + \hat{\mathbf{y}}_B\right),$$

whose value is invariant across the two states. Under the natural-unit benchmark, the resulting cash settlement requirement is lower than the direct interbank settlement requirement when $\rho > 0$. At date 1, the labourer can request cash from its bank. The

bank obtains cash from the central bank in exchange for a claim on its future output denominated in $\hat{\mathbf{y}}_m$, rather than supporting a direct interbank transfer through a claim denominated in the other bank's unit of account. Cash is accepted by artisans in both regions. A retail central bank digital currency could play a similar role if it were accepted as a means of payment across regions, as cash is assumed to be here.

The cash channel gives the bank an alternative way to support cross-region trade when direct interbank transfer is costly. The cash channel modifies the cross-region promise-keeping condition but does not replace the own-bank promise-keeping constraint in equation (9). If bank j allows either direct interbank transfers or cash withdrawals, the relevant cross-region condition becomes

$$n_j^{\varphi} \geq B_j \min\left\{(1-\chi_{j,k})\,\epsilon_{j,k}, \epsilon_{j,m}\right\}, \tag{41}$$

where $\epsilon_{j,m}$ is the settlement requirement for claims denominated in $\hat{\mathbf{y}}_m$, defined as $\epsilon_{j,m} \equiv \max_{\mathbf{p}\in\mathbf{P}} \mathbf{p}'\hat{\mathbf{y}}_m/\mathbf{p}'\hat{\mathbf{y}}_j$.

Proposition 9 (Cash as Settlement Channel)**.** *In the symmetric two-state model with cash denominated in* $\hat{\mathbf{y}}_m = \frac{1}{2}(\hat{\mathbf{y}}_A + \hat{\mathbf{y}}_B)$*:*

(a) *Cash caps the cross-region settlement requirement at*

$$\epsilon_{j,m} = \frac{1}{1-\rho},$$

which, under the natural unit of account, is strictly less than $\epsilon_{j,k} = (1+\rho)/(1-\rho)$ *for all* $\rho > 0$*.*

(b) *Under the natural unit of account, cash is an optimal option for every* $\rho \leq \rho_c$*, where* ρ_c *is defined in equation* (34)*, and is strictly preferred to direct interbank transfer for* $\rho_0 < \rho \leq \rho_c$*.*

(c) *Under the natural unit of account, cash weakly raises the autarky threshold to*

$$\rho_c^{(2)} = \max\{\rho^{(2)}, \rho_m^{(2)}\},$$

where $\rho_m^{(2)}$ *is the cash/autarky indifference threshold characterised in Appendix E. At* $\rho = \rho_c^{(2)}$*, autarky and cross-region trade are equally optimal.*

(d) *If* $\hat{\mathbf{b}}_k = \hat{\mathbf{y}}_m$*, then* $\epsilon_{j,k} = \epsilon_{j,m}$ *and bank* j *gains no additional benefit from the cash channel. Bank* k *can settle using cash at* $\epsilon_{k,m}$ *rather than direct interbank transfer at* $\epsilon_{k,j}$*.*

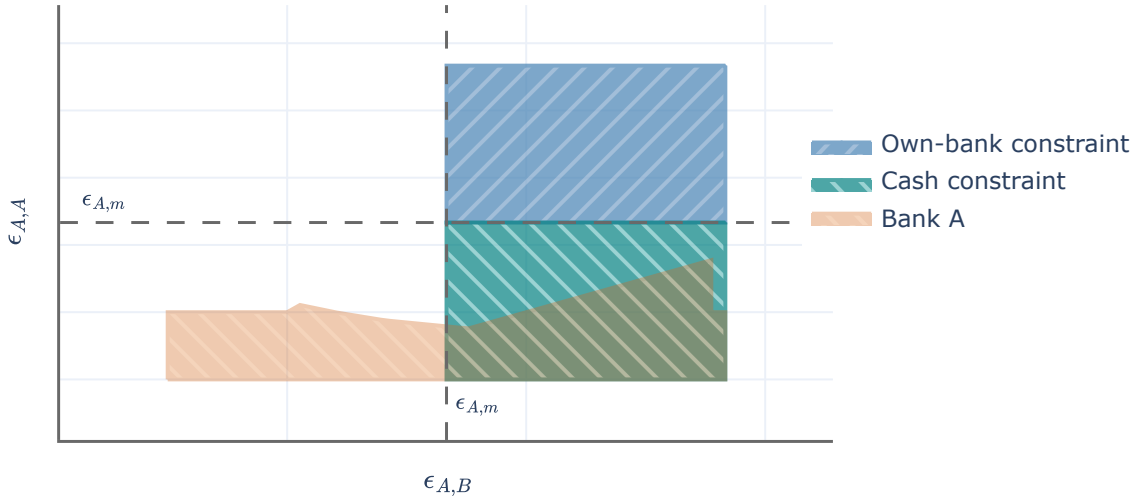


Figure 4: Bank A's best responses and cash settlement constraints. The horizontal axis reports the cross-bank settlement requirement $\epsilon_{A,B}$, while the vertical axis reports Bank A's own settlement requirement $\epsilon_{A,A}$. The figure illustrates investment and contract choices when the cash settlement requirement satisfies $\epsilon_{A,m} = 1/(1-\rho)$, with $\rho = 0.7$ in the symmetric two-state model without reserves. The figure is schematic.

Proof. See Appendix E. □

Under the natural-unit benchmark, cash is an optimal fee-free option for every $\rho \leq \rho_c$ and raises the autarky threshold to $\rho_c^{(2)}$ when $\rho_m^{(2)} > \rho^{(2)}$. The efficient-equilibrium existence boundary remains unchanged because the own-bank promise-keeping constraint continues to apply. When banks use the common monetary basket as their unit of account, the relevant own-bank settlement requirement is $\epsilon_{j,j} = 1/(1-\rho)$, which the cash channel does not relax.

The cash channel can also matter asymmetrically when a new entrant chooses a different unit of account from the incumbent. Suppose bank k adopts the monetary unit, so that $\hat{\mathbf{b}}_k = \hat{\mathbf{y}}_m$, while bank j chooses a different unit of account. For bank j, the direct obligation denominated in bank k's unit of account has the same settlement requirement as the cash claim, so cash provides no additional reduction in its settlement burden. For bank k, cash lowers the settlement requirement when $\epsilon_{k,m} < \epsilon_{k,j}$. In that case, cash can reduce the effect of bank j's unit-of-account choice on bank k's cross-region settlement burden. Bank k reaches the constrained-optimal investment level through cash settlement only if both its own promise-keeping requirement and the cash settlement requirement are at most $1/\varphi$.

Figure 4 illustrates the distinction between the own-bank and cash settlement constraints for Bank A, with $\rho = 0.7$ and $\epsilon_{A,m} = 1/(1-\rho)$. The shaded regions show where cash has

a lower settlement requirement than bank transfers. In the lower region, Bank A's own promise-keeping requirement is below the cash requirement, so cash settlement limits the contract. In the upper region, the own-bank requirement is higher, so switching to cash does not remove that constraint. The overlaid Bank A best-response correspondence shows the contracts chosen without cash.

5.4 Interaction of Policy Tools

Reserves and cash affect different margins of the problem. Reserves pool part of each bank's production risk through the monetary basket, reducing the effective relative-price dispersion under the post-policy natural-unit benchmark. Cash instead provides an alternative public settlement claim when direct interbank transfer is costly. The two instruments therefore affect different settlement conditions.

An efficient equilibrium exists over a wider range of relative-price dispersion when reserves are available, whereas the cash option leaves this boundary unchanged. Without reserves, an efficient equilibrium requires $\rho \leq 1 - \varphi$. With reserve requirement τ, this condition becomes $\rho(1-\tau) \leq 1 - \varphi$, or equivalently $\tau \geq \tau^{eff}$. Reserves can therefore support the existence of an efficient equilibrium at higher values of ρ. Cash does not have this effect because it relaxes the cross-region settlement condition without changing the banks' own promise-keeping constraints.

The two instruments are partial substitutes under the post-policy natural-unit benchmark. Without cash, a zero-fee outcome requires $\tau \geq \tau^*$. With cash, the relevant cross-region settlement condition uses $\epsilon_{j,m}(\tau) = 1/(1 - \tilde{\rho}(\tau))$ rather than the direct interbank settlement requirement $\epsilon_{j,k}(\tau)$. A zero-fee outcome is therefore guaranteed if $\tilde{\rho}(\tau) \leq \rho_c$, or equivalently if

$$\tau \;\geq\; \tau_c^* \;\equiv\; \max\left\{0,\; 1 - \frac{\rho_c}{\rho}\right\}. \tag{42}$$

Since $\rho_c > \rho^{(0)}$, it follows that $\tau_c^* < \tau^*$ whenever both reserve thresholds are interior. The reserve requirement that guarantees a zero-fee outcome is therefore lower when the cash channel is available under the post-policy natural-unit benchmark.

When $\rho \leq \rho_c$, a zero-fee outcome is guaranteed under the post-policy natural-unit benchmark without a positive reserve requirement. For $\rho > \rho_c$, the cash channel lowers the reserve requirement that guarantees a zero-fee outcome from τ^* to τ_c^*. The intermediate range $\rho^{(0)} < \rho \leq \rho_c$ therefore consists of cases in which introducing cash removes the need for a direct interbank transfer fee.

5.5 New Payment Intermediaries

The model illustrates how differences in repayment promises or issuer resources can increase settlement burdens, whether they arise between incumbent banks or through the introduction of a new payment intermediary. In the symmetric two-state natural-unit benchmark, the cutoffs $\rho^{(0)}$, $\rho^{(1)}$, and $\rho^{(2)}$ depend on relative-price dispersion rather than the aggregate price level, consistent with Doepke and Schneider (2017). As ρ increases, investment becomes constrained, transfer fees become positive, and banks eventually choose autarky. The model therefore links differences across issuers to progressively tighter limits on cross-issuer circulation.

When the banks are interpreted as issuers of USD stablecoins, their liabilities may share a nominal dollar denomination while differing in their effective denominations or issuer resources. Each directed settlement requirement depends on the promised basket relative to the settling issuer's resources over the relevant price support. A foreign-currency-backed stablecoin operating alongside domestic deposit banks provides one possible application. The same mechanism can apply to incumbent banks whose repayment promises or resource compositions differ sufficiently.

6. Information Sensitivity and Singleness

The policy analysis in Section 5 retains the baseline assumption that artisans value debt using its expected payoff. Labourers and artisans are symmetrically informed about the distribution of prices, and an artisan's participation condition is based on the expected value of the liability offered in exchange. This section instead requires the terms of exchange to remain acceptable under the lowest possible realised value of that liability. The extension changes the artisan's participation condition while leaving the baseline settlement and promise-keeping constraints unchanged. The formal analysis uses the symmetric two-state setting of Section 4.

The extension imposes one additional requirement. A labourer must set the terms of trade so that the artisan is willing to accept the offered debt for every possible realisation of its value. This is an ex ante worst-state requirement: the model does not introduce a signal that reveals the date-2 price state before the date-1 exchange. The requirement captures the idea that debt circulates without valuation costs only when its value is insensitive to the underlying state (Gorton and Pennacchi 1990; Gorton 2020; Dang, Gorton, and Holmström 2020). In the baseline environment, the artisan's participation condition uses

the liability's expected value. Under the extension, the labourer must instead compensate the artisan for its minimum possible value.

The relevant object is the shortfall in the liability's worst-state value relative to its normalised expected value. I refer to this as the *information sensitivity* of bank j's debt:

$$\sigma_j\left(\hat{\mathbf{b}}_j\right) = 1 - \min_{\mathbf{p}\in\mathbf{P}} \mathbf{p}'\hat{\mathbf{b}}_j. \tag{43}$$

Given the normalisation $\mathbb{E}[\mathbf{p}'\hat{\mathbf{b}}_j] = 1$, it follows that $\sigma_j \in [0, 1)$. When $\sigma_j = 0$, the liability has the same per-unit value throughout the price support. When $\sigma_j > 0$, its minimum value is $1 - \sigma_j < 1$. The statistic measures a worst-state payoff shortfall. It is not a probability of rejection, default, or a measure of opacity.

In the symmetric two-state model of Section 4, bank A's unit of account can be parameterised as $\hat{\mathbf{b}}_A = [\omega,\, 1-\omega]'$ with $\omega \in [0, 1]$. Substituting this parameterisation into equation (43) gives

$$\sigma_A(\omega) = \rho\,|1 - 2\omega|\,. \tag{44}$$

This expression is minimised at the symmetric midpoint $\omega = 1/2$, whose per-unit value is invariant across the two states. It is maximised at the corner baskets when $\rho > 0$.

Consider a labourer in region j who meets an artisan in region k and transfers a fraction $s_{j,k} \in [0, 1]$ of their debt in exchange for x_k units of the customised good. In a cross-region transaction, bank k issues its own liability to the artisan after receiving the interbank claim from bank j. The artisan's participation condition therefore depends on the worst-state value of bank k's debt:

$$x_k \le s_{j,k}(1 - \chi_{j,k})(1 - \sigma_k)\, B_j.$$

Conditional on trade being beneficial, the labourer transfers all available debt, so $s_{j,k} = 1$ and

$$x_k = (1 - \chi_{j,k})(1 - \sigma_k)B_j.$$

This acceptance condition is distinct from the state-contingent promise-keeping constraints faced by the banks.

Local acceptance instead depends on σ_j, since a local artisan receives bank j's debt. Let $g_L = (1+\gamma)(1-\sigma_j)$, $g_F = (1+\gamma)(1-\sigma_k)$, $u_L = 1 - \chi_{j,j}$, $u_F = 1 - \chi_{j,k}$ and $r = 1 - \alpha - \beta$, and let $I_L, I_F \in \{0, 1\}$ indicate local and cross-region trade, which occurs in channel $h \in \{L, F\}$ when permitted and $g_h u_h \ge 1$. Bank j chooses its unit of account, fees, n_j and B_j to maximise $n_j^{\varphi} - cB_j$ subject to $A\,B_j \ge n_j$, equation (9) and, if $I_F = 1$, equation (10), where $A = r + \alpha[I_L g_L u_L + 1 - I_L] + \beta[I_F g_F u_F + 1 - I_F]$ and c is A evaluated

at $g_L = g_F = 1$. Let $V^I(\rho)$ denote the value of this problem with $I_F = 0$, the optimised autarky payoff under worst-state acceptance.

Proposition 10 (Information-Sensitive Equilibrium)**.** *Consider the symmetric two-state model with $\rho_A = \rho_B = \rho$, and suppose the worst-state acceptance requirement holds. Let $\hat{\rho} \equiv (1-\varphi)(1-\beta)H/[(1-\varphi)(1-\beta)H + \varphi r]$, and call the profile in which both banks adopt the midpoint $\omega = \frac{1}{2}$, charge zero fees and choose optimal n_j and B_j the midpoint profile.*

(i) If $0 < \rho \leq 1-\varphi$, the midpoint profile is a Nash equilibrium with $\sigma_j = 0$ that attains the constrained-efficient benchmark W^*, and each bank's midpoint contract is its unique best response to the other's.

(ii) If $1-\varphi < \rho < 1$, the midpoint profile is a Nash equilibrium if and only if $\rho \leq \hat{\rho}$ and $W_m(\rho) \geq V^I(\rho)$, where W_m is defined in Appendix E. In this equilibrium both settlement constraints bind, $n_j = [H(1-\rho)]^{1/(1-\varphi)}$, $B_j = n_j/H$ and each bank earns $W_m(\rho)$; if $W_m(\rho) = V^I(\rho)$, autarky is equally optimal.

Proof. See Appendix F. □

The threshold $1-\varphi$ is the boundary above which the risk-free diversified unit of account can no longer attain W^*. Above it, both settlement constraints bind at the midpoint; moving towards a bank's own production basket relaxes equation (9) but raises σ_j, lowering local acceptance, and the midpoint survives this trade-off only for $\rho \leq \hat{\rho}$. In the baseline two-state model, the same threshold also marks the boundary below which efficient common-unit-of-account equilibria exist, as established in Proposition 6(a). At $\rho = 0$, the extension coincides with the baseline, so every denomination pair, with optimal contracts, supports an equilibrium attaining W^*. Outside the stated midpoint-equilibrium domain, equilibria need not be unique or symmetric; no complete characterisation is provided.

In the baseline model, bank profit is independent of the unit of account once the feasibility constraint is satisfied. Information sensitivity reduces that indifference: when local trade is active, a best response either has $\sigma_j = 0$ or makes equation (9) bind. Appendix F provides the formal argument.

Information sensitivity also strengthens the role of central bank reserves. When $\rho > 0$ and the reserve requirement satisfies

$$\tau \geq \frac{\rho + \varphi - 1}{\rho},$$

effective price dispersion falls to at most $1 - \varphi$. At that point, the midpoint profile is again an equilibrium with $\sigma_j = 0$ that attains W^*. The same condition places the economy within the efficient-equilibrium region identified in Proposition 8(a). Appendix F formalises this result.

Each bank's unit of account now affects three objectives, only two of which enter its own payoff. Borrowing capacity is highest when $\epsilon_{j,j}$ is low, which favours a unit of account close to the bank's own output. Interoperability is greatest when $\epsilon_{k,j}$ is low, which favours a unit of account close to bank k's output. Information sensitivity is minimised when $\sigma_j = 0$, which favours the risk-free diversified bundle. When $\rho > 0$, no feasible unit of account achieves all three objectives at once.

Cash denominated in the monetary unit of account $\hat{\mathbf{y}}_m = \frac{1}{2}(\hat{\mathbf{y}}_A + \hat{\mathbf{y}}_B)$ has information sensitivity $\sigma_m = 0$, since $\mathbf{p}'\hat{\mathbf{y}}_m = 1$ in every state. When bank debt is information-sensitive, labourers face lower effective gains from trade. Cash does not incur this information-sensitivity discount because its payoff is state-invariant. This property is distinct from the cash settlement requirement $\epsilon_{j,m}$ analysed in Section 5.

7. Conclusion

Fragmentation can arise when liabilities issued within the same monetary system have different effective denominations and their issuers rely on different resources to honour them. When each bank denominates debt in its own unit of account, a transferred claim may be especially valuable relative to the settling bank's resources in some states. The resulting settlement exposure can reduce investment. As it increases, transfer fees become positive and, beyond a threshold, banks prohibit interbank transfers, leaving private money non-interoperable across issuers.

Coordination on a shared unit of account extends the range over which fee-free exchange can occur. Such coordination supports efficient exchange when the shared repayment basket is sufficiently well aligned with issuer resources. As this alignment weakens, a common-unit equilibrium may remain feasible while investment becomes constrained. Efficient exchange therefore does not generally require parity across forms of private money. Nor does parity guarantee efficient exchange. Efficiency instead requires settlement exposures to remain low enough for banks to support the constrained-optimal level of investment.

The policy analysis shows that public instruments affect these settlement conditions through different mechanisms. Central bank reserves provide a common settlement asset

and pool part of the state-contingent variation in issuer resources. They can support efficient, fee-free exchange even when banks have substantially different asset bases. Cash provides an alternative settlement channel through central bank money and can support interoperability when direct settlement across issuers is costly.

A common-unit-of-account mandate removes the additional settlement exposure created by differences in effective denomination, but does not remove banks' own promise-keeping constraints. If the mandated repayment basket is poorly aligned with issuer resources, the mandate can constrain investment even though the liabilities are valued at par. Mandating a common effective denomination is therefore not equivalent to restoring efficiency.

The broader implication is that a common nominal currency label does not by itself preserve monetary singleness. Two stablecoins, tokenised deposits, or other private liabilities may share a nominal denomination while differing in their effective denominations and the resources available to their issuers. These differences matter when liabilities circulate across issuers and create state-contingent settlement obligations. Limited departures from par can be consistent with efficient exchange, whereas larger settlement exposures can constrain investment and undermine interoperability.

When acceptance is information-sensitive, variation in the realised value of private liabilities provides a further channel through which differences in effective denomination reduce the gains from trade. This extension broadens the analysis to include acceptance while retaining state-by-state promise keeping and the absence of default. In practice, singleness also depends on confidence in issuers' ability to honour their obligations and on the conditions governing redemption. This paper focuses on the settlement burdens that can persist even when repayment is assured, and shows how central bank money can support circulation across issuers.

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Appendix

A. Foundational Bank Optimisation

Bank j's Problem without Interbank Transfers

The Lagrangian is

$$\begin{aligned}\mathcal{L}_j^{(0)} =& n_j^{\varphi} - (1-\alpha\chi_{j,j})B_j \\ &+ \lambda_N\left[((1+\gamma)\alpha(1-\chi_{j,j})+1-\alpha)B_j - n_j\right] \\ &+ \mu_j^{+}\left[(1+\gamma)(1-\chi_{j,j})-1\right] + \mu_j^{-}\chi_{j,j}\end{aligned} \tag{A.1}$$

I take the first-order conditions with respect to $n_j, B_j, \chi_{j,j}$, which are as follows:

$$\frac{\partial \mathcal{L}_j^{(0)}}{\partial n_j}: \quad \varphi n_j^{\varphi-1} - \lambda_N = 0 \tag{A.2}$$

$$\frac{\partial \mathcal{L}_j^{(0)}}{\partial B_j}: \quad \lambda_N\left[(1+\gamma)\alpha(1-\chi_{j,j})+1-\alpha\right] - (1-\alpha\chi_{j,j}) = 0 \tag{A.3}$$

$$\frac{\partial \mathcal{L}_j^{(0)}}{\partial \chi_{j,j}}: \quad \alpha B_j - \lambda_N(1+\gamma)\alpha B_j - \mu_j^{+}(1+\gamma) + \mu_j^{-} = 0 \tag{A.4}$$

It follows from the latter equation that

$$\chi_{j,j} = 0 \iff \lambda_N > \frac{1}{1+\gamma} \tag{A.5}$$

Rearranging the second equation yields

$$\lambda_N = \frac{1-\alpha\chi_{j,j}}{(1+\gamma)\alpha(1-\chi_{j,j})+1-\alpha} \tag{A.6}$$

where I note that this implies $\lambda_N > \frac{1}{1+\gamma}$ for every feasible $\chi_{j,j} \in [0, \gamma/(1+\gamma)]$. Thus I note that it is optimal in autarky for firms not to charge for intrabank transfers.

Substituting out λ_N in the first equation yields an equation for the quantity of labour hired in autarky

$$n_j^{(0)} = \left[\varphi\left(1+\gamma\alpha\right)\right]^{\frac{1}{1-\varphi}}. \tag{A.7}$$

The loan size follows from the budget constraint

$$B_j^{(0)} = \varphi^{\frac{1}{1-\varphi}} (1+\gamma\alpha)^{\frac{\varphi}{1-\varphi}}, \tag{A.8}$$

and the profit to bank j is

$$V_j^{(0)} = \varphi^{\frac{\varphi}{1-\varphi}} (1+\gamma\alpha)^{\frac{\varphi}{1-\varphi}} (1-\varphi). \tag{A.9}$$

Finally, note that the bank's unit of account can be set to any value that ensures the promise keeping condition described by equation (3) holds. This is satisfied so long as

$$\epsilon_{j,j} \leq \frac{1}{\varphi}. \tag{A.10}$$

Bank j's Problem with Interbank Transfers

I first solve the relaxed problem that omits the own-bank promise-keeping constraint, equation (9); Lemma C.1 characterises the full optimum, including when equation (9) binds. The Lagrangian then becomes

$$\begin{aligned}
\mathcal{L}_j =& n_j^{\varphi} - (1-\alpha\chi_{j,j} - \beta\chi_{j,k})B_j \\
&+ \lambda_N \left[((1+\gamma)(\alpha(1-\chi_{j,j}) + \beta(1-\chi_{j,k})) + 1 - \alpha - \beta)B_j - n_j\right] \\
&+ \lambda_k \left[\frac{n_j^{\varphi}}{\epsilon_{j,k}} - (1-\chi_{j,k})B_j\right] + \mu_j^{+}\left[(1+\gamma)(1-\chi_{j,j}) - 1\right] \\
&+ \mu_k^{+}\left[(1+\gamma)(1-\chi_{j,k}) - 1\right] + \mu_j^{-}\chi_{j,j} + \mu_k^{-}\chi_{j,k}
\end{aligned} \tag{A.11}$$

I take the first-order conditions with respect to $n_j, B_j, \chi_{j,j}, \chi_{j,k}$, which are as follows:

$$\frac{\partial \mathcal{L}_j}{\partial n_j}: \quad \varphi n_j^{\varphi-1}\left(1 + \frac{1}{\epsilon_{j,k}}\lambda_k\right) - \lambda_N = 0 \tag{A.12}$$

$$\begin{aligned}
\frac{\partial \mathcal{L}_j}{\partial B_j}: \quad & \lambda_N\left[(1+\gamma)(\alpha(1-\chi_{j,j}) + \beta(1-\chi_{j,k})) + 1 - \alpha - \beta\right] \\
& - (1 - \alpha\chi_{j,j} - \beta\chi_{j,k}) - \lambda_k(1-\chi_{j,k}) = 0
\end{aligned} \tag{A.13}$$

$$\frac{\partial \mathcal{L}_j}{\partial \chi_{j,j}}: \quad \alpha B_j - \lambda_N(1+\gamma)\alpha B_j - \mu_j^{+}(1+\gamma) + \mu_j^{-} = 0 \tag{A.14}$$

$$\frac{\partial \mathcal{L}_j}{\partial \chi_{j,k}}: \quad \beta B_j - \lambda_N(1+\gamma)\beta B_j + \lambda_k B_j - \mu_k^{+}(1+\gamma) + \mu_k^{-} = 0 \tag{A.15}$$

B. Proofs of Main Equilibrium Results

Proof of Proposition 2

Case (i): $\epsilon_{j,k} \leq 1/\varphi$

First assume that $\lambda_k = 0$ and later verify under which conditions this holds.

From rearranging equation (A.14) it follows that

$$\chi_{j,j} = 0 \iff \lambda_N > \frac{1}{1+\gamma}. \tag{B.1}$$

Similarly, rearranging equation (A.15) yields

$$\chi_{j,k} = 0 \iff \lambda_N > \frac{\beta + \lambda_k}{(1+\gamma)\beta}. \tag{B.2}$$

Note that in the case where $\lambda_k = 0$ equations (B.1) and (B.2) are equivalent.

Rearranging equation (A.13) yields the following equation for λ_N

$$\lambda_N = \frac{1 - \alpha\chi_{j,j} - \beta\chi_{j,k} + \lambda_k(1-\chi_{j,k})}{1 + \gamma\,(\alpha+\beta) - (1+\gamma)\,(\alpha\chi_{j,j} + \beta\chi_{j,k})}, \tag{B.3}$$

and note that in the case where $\lambda_k = 0$, equations (B.1) and (B.2) imply that $\chi_{j,j} = \chi_{j,k} = 0$.

Now substituting out λ_N from equation (A.12) and rearranging yields

$$n_j = \left[\varphi\,(1 + \gamma(\alpha+\beta))\right]^{\frac{1}{1-\varphi}}. \tag{B.4}$$

In equilibrium, the labourer's participation constraint, equation (24), holds with equality, so $B_j = n_j/(1+\gamma(\alpha+\beta))$; substituting n_j and B_j into bank j's profit function gives the debt size $B_j^{(1)}$ and profit $W_j^{(1)}$ stated in Proposition 2(i). As $n_j^{\varphi}/B_j = 1/\varphi$, the bank's initial promise-keeping constraint, equation (9), is satisfied if and only if $\epsilon_{j,j} \leq 1/\varphi$.

Finally, it remains to find the conditions under which $\lambda_k = 0$. For this to hold it must be the case that the interbank promise-keeping constraint, equation (10) does not reduce investment. Substituting the equations for n_j and B_j into equation (10) yields the following inequality

$$\epsilon_{j,k} \leq \frac{1}{\varphi}. \tag{B.5}$$

Case (ii): $\epsilon_{j,k} > 1/\varphi$

Consider now the case where $\epsilon_{j,k} > \frac{1}{\varphi}$ and thus $\lambda_k > 0$. Here, equation (B.1) is still satisfied and $\chi_{j,j} = 0$.

Rearranging the first-order conditions and binding constraints yields the following system of equations:

$$\lambda_N = \frac{1 - \beta\chi_{j,k} + (1 - \chi_{j,k})\lambda_k}{1 + \gamma\alpha + \beta(\gamma - (1+\gamma)\chi_{j,k})} \tag{B.6}$$

$$n_j^{\varphi-1}\frac{1}{\epsilon_{j,k}}(1 + \gamma\alpha + \beta(\gamma - (1+\gamma)\chi_{j,k})) = (1 - \chi_{j,k}) \tag{B.7}$$

and

$$\lambda_N = \varphi n_j^{\varphi-1}\left(1 + \lambda_k\frac{1}{\epsilon_{j,k}}\right). \tag{B.8}$$

Solving this system of equations for n_j yields

$$n_j = \left[\frac{1 + \gamma\alpha + \beta\left(\gamma - (1+\gamma)\chi_{j,k}\right)}{\epsilon_{j,k}\left(1 - \chi_{j,k}\right)}\right]^{\frac{1}{1-\varphi}} \tag{B.9}$$

The debt level follows from rearranging equation (24)

$$B_j = \frac{n_j}{1 + \gamma\alpha + \beta(\gamma - (1+\gamma)\chi_{j,k})} \tag{B.10}$$

which yields

$$B_j = \left(\frac{1}{\epsilon_{j,k}\left(1 - \chi_{j,k}\right)}\right)^{\frac{1}{1-\varphi}}\left[1 + \gamma\alpha + \beta\left(\gamma - (1+\gamma)\chi_{j,k}\right)\right]^{\frac{\varphi}{1-\varphi}}. \tag{B.11}$$

Solving for bank j's profit yields

$$W_j = \left[\frac{1 + \gamma\alpha + \beta\left(\gamma - (1+\gamma)\chi_{j,k}\right)}{\epsilon_{j,k}\left(1 - \chi_{j,k}\right)}\right]^{\frac{\varphi}{1-\varphi}}\left(1 - \frac{1 - \beta\chi_{j,k}}{\epsilon_{j,k}\left(1 - \chi_{j,k}\right)}\right) \tag{B.12}$$

These expressions use only the binding participation constraint and equation (10), so they hold at the optimal fee $\chi^*_{j,k}$ of Lemma C.1, including where equation (9) also binds; by its proof, $n_j^{\varphi}/B_j = \epsilon_{j,k}(1 - \chi^*_{j,k}) \geq 1$, so any unit of account with $\epsilon_{j,j} \leq n_j^{\varphi}/B_j$, including the natural unit of account, implements the contract. If $1/\varphi < \epsilon_{j,k} < \epsilon'_{j,k}$, the optimal contract at $\epsilon'_{j,k}$ satisfies equation (10) strictly at $\epsilon_{j,k}$ and no other constraint involves $\epsilon_{j,k}$; since the unique optimal contract at $\epsilon_{j,k}$ has equation (10) binding, $W_j(\epsilon_{j,k}) > W_j(\epsilon'_{j,k})$.

Proof of Proposition 3

When $\epsilon_{j,k} = 1/\varphi$, case (i) of Proposition 2 gives $W_j^{(1)} > V_j^{(0)}$ since $\beta > 0$ and $\gamma > 0$ imply that the gains from interbank trade are strictly positive. Since $\lim_{\epsilon_{j,k}\to\infty} W_j(\epsilon_{j,k}) = 0 < V_j^{(0)}$ and W_j is continuous and strictly decreasing for $\epsilon_{j,k} > 1/\varphi$ (by Proposition 2(ii)), the intermediate value theorem yields a unique crossing point $\bar{\epsilon}_{j,k}^{(2)}$. For the comparative statics, $(1-\varphi)(1-\alpha-\beta) < 1+\gamma\alpha-\beta$ gives $\varphi\bar{\epsilon}_{j,k}^{(0)} < 1+\beta\gamma$, so both terms defining $\bar{\epsilon}_{j,k}^{(1)}$ in Lemma C.1 are below $(1+\gamma)/\varphi$. For $\epsilon_{j,k} \geq \bar{\epsilon}_{j,k}^{(1)}$, substituting $\chi_{j,k}^* = \gamma/(1+\gamma)$ into case (ii) of Proposition 2 and comparing with $V_j^{(0)}$ in Proposition 1 shows that, with $z = \varphi\epsilon_{j,k}/(1+\gamma)$, $W_j \geq V_j^{(0)}$ if and only if $g(z) \equiv (1-\varphi)z^{\frac{1}{1-\varphi}} - z + \varphi - \varphi\beta\gamma/(1+\gamma) \leq 0$. As $g(1) < 0$, $W_j > V_j^{(0)}$ at $\epsilon_{j,k} = (1+\gamma)/\varphi$, so $\bar{\epsilon}_{j,k}^{(1)} < (1+\gamma)/\varphi < \bar{\epsilon}_{j,k}^{(2)}$ and $\bar{z} = \varphi\bar{\epsilon}_{j,k}^{(2)}/(1+\gamma) > 1$ solves $g(\bar{z}) = 0$. Since $g'(\bar{z}) = \bar{z}^{\frac{\varphi}{1-\varphi}} - 1 > 0$ and g is strictly decreasing in β and γ and independent of α, $\bar{z}$ is strictly increasing in β and γ and independent of α. Hence $\bar{\epsilon}_{j,k}^{(2)} = (1+\gamma)\bar{z}/\varphi$ is strictly increasing in β and γ, and independent of α.

Proof of Proposition 4

Autarky. As $n_i^{\varphi}/B_i = 1/\varphi$ at the contract of Proposition 1, autarky yields $V_i^{(0)}$ if and only if $\epsilon_{i,i} \leq 1/\varphi$, and it is then a best response if and only if $W_i(\epsilon_{i,\ell}) \leq V_i^{(0)}$, that is, $\epsilon_{i,\ell} \geq \bar{\epsilon}_{i,\ell}^{(2)}$ by Proposition 3.

Conditional contract regimes. Fix a feasible pair of units of account, equivalently a feasible pair $(\epsilon_{j,k}, \epsilon_{k,j})$. Bank i's contract problem depends on bank ℓ's unit of account only through $\epsilon_{i,\ell}$. Combining Propositions 2–3 and Lemma C.1 yields the four regimes listed in the proposition. The intervals are mutually exclusive and exhaustive conditional on the settlement requirement faced by bank i.

Efficient fee-free equilibrium. If $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k = \hat{\mathbf{b}}$ with $\max_{\mathbf{p}\in\mathbf{P}} \mathbf{p}'\hat{\mathbf{b}}/\mathbf{p}'\hat{\mathbf{y}}_i \leq 1/\varphi$ for both banks, then the conditions for case (i) of Proposition 2 are met simultaneously. Both banks set zero transfer fees, investment is at the constrained optimum, and the resulting profile is a Nash equilibrium.

C. Fee Cutoffs

Lemma C.1 (Fee Cutoffs)**.** *Suppose that bank j allows interbank transfers and bank k's unit of account induces a settlement requirement $\epsilon_{j,k} > \frac{1}{\varphi}$. Then there exist two cutoffs*

$$\bar{\epsilon}_{j,k}^{(0)} = \frac{1}{\varphi}\left(1 + \beta\left(\frac{\gamma(1-\varphi)(1-\alpha-\beta)}{1+\gamma\alpha-\beta}\right)\right)$$

and

$$\bar{\epsilon}_{j,k}^{(1)} = \max\left\{\bar{\epsilon}_{j,k}^{(0)} + \frac{\gamma(1-\beta)}{\varphi},\ 1+\gamma\right\}$$

where $\frac{1}{\varphi} < \bar{\epsilon}_{j,k}^{(0)} < \bar{\epsilon}_{j,k}^{(1)}$ *such that*

$$\chi_{j,k}^{*}(\epsilon_{j,k}) = 0 \iff \epsilon_{j,k} \leq \bar{\epsilon}_{j,k}^{(0)}$$

and

$$\chi_{j,k}^{*}(\epsilon_{j,k}) = \frac{\gamma}{1+\gamma} \iff \epsilon_{j,k} \geq \bar{\epsilon}_{j,k}^{(1)}$$

and, for $\bar{\epsilon}_{j,k}^{(0)} < \epsilon_{j,k} < \bar{\epsilon}_{j,k}^{(1)}$,

$$\chi_{j,k}^{*}(\epsilon_{j,k}) = \min\left\{\frac{\varphi\left(\epsilon_{j,k} - \bar{\epsilon}_{j,k}^{(0)}\right)}{1-\beta+\varphi\left(\epsilon_{j,k} - \bar{\epsilon}_{j,k}^{(0)}\right)},\ 1 - \frac{1}{\epsilon_{j,k}}\right\}.$$

Proof. Consider equation (A.15) in the case where $\lambda_k > 0$.

When equation (10) and the participation constraint bind, equations (A.12) and (A.13) give

$$\lambda_k = \frac{1}{1-\varphi}\left(\varphi\epsilon_{j,k} - \frac{1-\beta\chi_{j,k}}{1-\chi_{j,k}}\right). \tag{C.1}$$

There are three possible scenarios to consider. First, consider the case where $\chi_{j,k} = 0$. This occurs if

$$\frac{1+\lambda_k}{1+\gamma(\alpha+\beta)} \geq \frac{\beta+\lambda_k}{(1+\gamma)\beta}, \tag{C.2}$$

which simplifies to

$$\lambda_k \leq \frac{\gamma\beta(1-\alpha-\beta)}{1-\beta+\gamma\alpha}. \tag{C.3}$$

At $\chi_{j,k} = 0$, equation (C.1) gives $\lambda_k = (\varphi\epsilon_{j,k} - 1)/(1-\varphi)$, and combining this with the last bound and rearranging yields the result that

$$\chi_{j,k} = 0 \iff \epsilon_{j,k} \leq \frac{1}{\varphi}\left(1 + \beta\left(\frac{\gamma(1-\varphi)(1-\alpha-\beta)}{1+\gamma\alpha-\beta}\right)\right) \equiv \bar{\epsilon}_{j,k}^{(0)}. \tag{C.4}$$

Similarly, consider the case where the maximum interbank transfer fee is charged, that is $\chi_{j,k} = \frac{\gamma}{1+\gamma}$. This occurs if

$$\frac{1 - \frac{\beta\gamma}{1+\gamma} + \frac{1}{1+\gamma}\lambda_k}{1+\gamma\alpha} \leq \frac{\beta+\lambda_k}{(1+\gamma)\beta}, \tag{C.5}$$

which simplifies to $\lambda_k \geq \gamma\beta(1-\alpha-\beta)/(1+\gamma\alpha-\beta)$.

At $\chi_{j,k} = \gamma/(1+\gamma)$, equation (C.1) gives $\lambda_k = (\varphi\epsilon_{j,k} - (1+\gamma) + \beta\gamma)/(1-\varphi)$, and combining this with that bound and rearranging yields the result that

$$\chi_{j,k} = \frac{\gamma}{1+\gamma} \iff \epsilon_{j,k} \geq \bar{\epsilon}^{(0)}_{j,k} + \frac{\gamma(1-\beta)}{\varphi}. \tag{C.6}$$

To verify the ordering $1/\varphi < \bar{\epsilon}^{(0)}_{j,k} < \bar{\epsilon}^{(1)}_{j,k}$, the first inequality follows immediately from the definition of $\bar{\epsilon}^{(0)}_{j,k}$, and the second from the definition of $\bar{\epsilon}^{(1)}_{j,k}$ since $\gamma(1-\beta)/\varphi > 0$.

If $\bar{\epsilon}^{(0)}_{j,k} < \epsilon_{j,k} < \bar{\epsilon}^{(0)}_{j,k} + \gamma(1-\beta)/\varphi$, both fee bounds are slack, and equations (A.13) and (A.15) imply $\lambda_k = \gamma\beta(1-\alpha-\beta)/(1+\gamma\alpha-\beta)$ for every interior fee; equating this with equation (C.1) and using equation (C.4) gives the first term of the minimum in the lemma, which lies in $(0, \gamma/(1+\gamma))$ on this interval.

Own-bank constraint and global optimum. The fee derived above, the relaxed fee, solves the problem of Appendix A, which omits equation (9). As $\epsilon_{j,j} \geq 1$, with equality under the natural unit of account (Section 2), some unit of account satisfies equation (9) if and only if $n_j^{\varphi} \geq B_j$; this adds only a nonnegative multiplier to the numerator of λ_N in equation (A.13), so $\chi_{j,j} = 0$ still. Fix $\chi_{j,k} = \chi$ and let $A_\chi = 1 + \gamma\alpha + \beta(\gamma - (1+\gamma)\chi)$. With participation binding, $B_j = n_j/A_\chi$, profit $n_j^{\varphi} - (1-\beta\chi)n_j/A_\chi$ is strictly concave in n_j, $(1-\beta\chi)/A_\chi$ is strictly increasing in χ, and $n_j^{\varphi} \geq B_j$ becomes $n_j^{1-\varphi} \leq A_\chi$.

In the relaxed problem, equation (10) binds exactly where the right-hand side of equation (C.1), which decreases in χ, is positive; at higher fees investment is unconstrained and profit falls in χ. Where it binds, the envelope theorem and equation (A.13) give a derivative of profit in χ equal to $(1+\gamma\alpha-\beta)B_j/A_\chi$ times $\lambda_k - \gamma\beta(1-\alpha-\beta)/(1+\gamma\alpha-\beta)$. Relaxed profit therefore strictly increases up to the relaxed fee and strictly decreases beyond it, and, since the three cases above give $\lambda_k > 0$ at the relaxed fee, equation (10) binds at every fee up to the relaxed fee.

With $n_j^{\varphi} \geq B_j$ retained, for $\chi \leq 1 - 1/\epsilon_{j,k}$ equation (10) implies it, so profit is unchanged; for $\chi \geq 1 - 1/\epsilon_{j,k}$ it implies equation (10), and a higher χ lowers A_χ and raises $(1-\beta\chi)/A_\chi$, so profit strictly falls. The optimal fee, and with it investment and debt, is therefore unique: the fee is the smaller of the relaxed fee and $1 - 1/\epsilon_{j,k} > 0$, equation (10) binds there, and $n_j^{\varphi}/B_j = \epsilon_{j,k}(1-\chi^*_{j,k}) \geq 1$, with equality, and equation (9) binding under the natural unit of account, when $\chi^*_{j,k} = 1 - 1/\epsilon_{j,k}$. Since $1 - 1/\epsilon_{j,k} \geq \gamma/(1+\gamma)$ if and only if $\epsilon_{j,k} \geq 1+\gamma$, this yields both cutoffs and, on $(\bar{\epsilon}^{(0)}_{j,k}, \bar{\epsilon}^{(1)}_{j,k})$, the stated minimum, which is continuous and strictly increasing in $\epsilon_{j,k}$; where the relaxed fee there equals $\gamma/(1+\gamma)$, $\epsilon_{j,k} < 1+\gamma$, so the first term exceeds $1 - 1/\epsilon_{j,k}$. □

D. Proofs for Two-State Results

Proof of Proposition 5

Proof. The mapping $\epsilon(\rho) = (1+\rho)/(1-\rho)$ is strictly increasing, so the ordering $\bar{\epsilon}^{(0)} < \bar{\epsilon}^{(1)} < \bar{\epsilon}^{(2)}$ established in the proofs of Lemma C.1 and Proposition 3 maps to $\rho^{(0)} < \rho^{(1)} < \rho^{(2)}$. Symmetry ($\epsilon_{A,B} = \epsilon_{B,A}$) implies both banks face identical optimisation problems, so $\chi^*_{A,B} = \chi^*_{B,A}$.

For case (a): when $\rho \leq \rho_0$, $\epsilon(\rho) \leq 1/\varphi$. Under the natural unit of account, $\chi^* = 0$ and investment is at the constrained optimum by Proposition 2(i). For any other equilibrium, bank k's unit of account satisfies $\epsilon_{j,k} = \max_s\{\mathbf{p}'_s\hat{\mathbf{b}}_k/\mathbf{p}'_s\hat{\mathbf{y}}_j\} \leq \max_s\{\mathbf{p}'_s\hat{\mathbf{y}}_k/\mathbf{p}'_s\hat{\mathbf{y}}_j\} = \epsilon(\rho) \leq 1/\varphi$, where the first inequality follows because, in the two-state model with specialised production, $\hat{\mathbf{y}}_k$ maximises $\max_s\{\mathbf{p}'_s\hat{\mathbf{b}}/\mathbf{p}'_s\hat{\mathbf{y}}_j\}$ over all normalised baskets $\hat{\mathbf{b}}$ with $\mathbf{p}'_s\hat{\mathbf{b}} \geq 0$. Since $\epsilon_{j,k} \leq 1/\varphi$ for both banks in every equilibrium, both banks allow transfers by Proposition 3, and all equilibria are efficient.

For case (b): when $\rho_0 < \rho \leq \rho^{(0)}$, $\epsilon(\rho) \leq \bar{\epsilon}^{(0)}$ ensures $\chi^* = 0$ under the natural unit of account by Lemma C.1, but $\epsilon(\rho) > 1/\varphi$ so investment is below the constrained optimum.

For case (c): when $\rho^{(0)} < \rho \leq \rho^{(2)}$, $\epsilon(\rho) > \bar{\epsilon}^{(0)}$ and bank j charges $\chi^*(\rho) > 0$ by Lemma C.1; at $\rho = \rho^{(2)}$, autarky is equally optimal by Proposition 3.

For case (d): when $\rho > \rho^{(2)}$, $\epsilon(\rho) > \bar{\epsilon}^{(2)}$ implies $W_j(\epsilon) < V_j^{(0)}$ by Proposition 3, so banks prefer autarky. □

Proof of Proposition 6

Proof. Suppose both banks adopt a common unit of account $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k = [\omega, 1-\omega]'$. Under a common unit of account, $\epsilon_{j,k} = \epsilon_{j,j}$ for each bank, so equation (9) implies equation (10) at every fee, and setting $\chi_{j,k} > 0$ relaxes only the redundant equation (10) while tightening the participation constraint; hence $\chi^*_{j,k} = 0$. If bank j deviates to $\hat{\mathbf{b}}_j = \hat{\mathbf{y}}_j$, its promise-keeping constraint relaxes to $\epsilon_{j,j} = 1$ while its interbank settlement cost $\epsilon_{j,k} = \max_s\{\mathbf{p}'_s[\omega, 1-\omega]'/\mathbf{p}'_s\hat{\mathbf{y}}_j\}$ is unchanged (bank k's basket is held fixed). The deviating bank can then set $\chi_{j,k} = \chi^*(\epsilon_{j,k}) > 0$ whenever $\epsilon_{j,k} > \bar{\epsilon}^{(0)}$, strictly increasing profit relative to the common-unit-of-account outcome. By symmetry, the most favourable common basket is $\omega = 1/2$, giving $\epsilon_{j,k} = 1/(1-\rho)$ for both banks. The condition $1/(1-\rho) > \bar{\epsilon}^{(0)}$ is equivalent to $\rho > \rho_c$. For $\rho \leq \rho_c$, deviation does not increase profit ($\chi^*(\epsilon_{j,k}) = 0$ under

deviation as well), nor does autarky, since $\epsilon_{j,k} \leq \bar{\epsilon}^{(0)} < \bar{\epsilon}^{(2)}$, so banks are indifferent and the common-unit-of-account profile is a Nash equilibrium. For $\rho > \rho_c$, deviation is strictly profitable at any common basket, so no common-unit-of-account Nash equilibrium exists. This establishes parts (b) and (c).

For part (a): under the common basket at $\omega = 1/2$, both banks face $\epsilon_{j,j} = 1/(1-\rho)$. By Proposition 4, efficient equilibria exist when $1/(1-\rho) \leq 1/\varphi$, i.e. $\rho \leq 1-\varphi$. Since $1-\varphi < \rho_c$ (as $\bar{\epsilon}^{(0)} > 1/\varphi$), the common-unit-of-account equilibrium exists and is efficient for $\rho \leq 1-\varphi$. □

Proof of Proposition 7

Proof. Part (a) follows immediately since zero-fee equilibria with $\epsilon(\rho) \leq 1/\varphi$ attain the constrained-efficient benchmark.

For part (b), substituting the equilibrium fee $\chi^*(\rho)$ into the bank's payoff function yields

$$\Delta W(\rho) = W^* - \left[\frac{1+\gamma\alpha+\beta\left(\gamma-(1+\gamma)\chi^*(\rho)\right)}{\epsilon(\rho)\left(1-\chi^*(\rho)\right)}\right]^{\frac{\varphi}{1-\varphi}} \left(1 - \frac{1-\beta\chi^*(\rho)}{\epsilon(\rho)\left(1-\chi^*(\rho)\right)}\right), \quad \text{(D.1)}$$

where $\chi^*(\rho) \equiv \chi^*(\epsilon(\rho))$ is the optimal fee from Lemma C.1 and $\epsilon(\rho) = (1+\rho)/(1-\rho)$. For the maximum-fee sub-case $\rho^{(1)} \leq \rho \leq \rho^{(2)}$, substituting $\chi^* = \gamma/(1+\gamma)$ yields

$$\Delta W(\rho) = W^* - \left[\frac{(1+\gamma\alpha)(1+\gamma)}{\epsilon(\rho)}\right]^{\frac{\varphi}{1-\varphi}} \left(1 - \frac{1-\beta\gamma/(1+\gamma)}{\epsilon(\rho)/(1+\gamma)}\right). \quad \text{(D.2)}$$

For part (c), both banks choose autarky by Proposition 5(d) and each earns $V_j^{(0)}$ from Proposition 1, so $\Delta W(\rho) = W^* - V_j^{(0)}$.

Monotonicity in ρ follows from Proposition 2(ii): W_j is strictly decreasing in $\epsilon_{j,k}$ for $\epsilon_{j,k} > 1/\varphi$, and $\epsilon(\rho)$ is strictly increasing in ρ. Welfare loss is therefore strictly increasing on the entire interval $(\rho_0, \rho^{(2)}]$. The autarky payoff $V_j^{(0)}$ does not depend on ρ, hence ΔW is constant for $\rho > \rho^{(2)}$. Continuity across the boundary at $\rho^{(1)}$ follows from continuity of $\chi^*(\rho)$. At $\rho^{(2)}$, $W_j(\bar{\epsilon}^{(2)}) = V_j^{(0)}$ by definition, so ΔW is continuous at $\rho^{(2)}$. □

Closed-Form Transfer Fee

In the symmetric two-state model, for $\rho^{(0)} < \rho < \rho^{(1)}$, substituting $\epsilon = (1+\rho)/(1-\rho)$ into the fee in Lemma C.1 yields

$$\chi^*(\rho) = \min\left\{\frac{\varphi\left(\epsilon(\rho) - \bar{\epsilon}^{(0)}\right)}{1-\beta+\varphi\left(\epsilon(\rho)-\bar{\epsilon}^{(0)}\right)},\ \frac{2\rho}{1+\rho}\right\}. \tag{D.3}$$

By Lemma C.1 and the definitions of $\rho^{(0)}$ and $\rho^{(1)}$, $\chi^*(\rho^{(0)}) = 0$ and $\chi^*(\rho^{(1)}) = \gamma/(1+\gamma)$, and $\chi^*(\rho)$ is strictly increasing on $(\rho^{(0)}, \rho^{(1)})$ because both terms in the minimum are.

E. Policy Proofs and Results

Proposition E.1 (Common Unit of Account Mandate)**.** *In the symmetric two-state model:*

(a) *Under any mandate $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k = \hat{\mathbf{y}}_a$, both banks optimally set $\chi^*_{j,k} = 0$.*

(b) *For $\rho \le \rho_c$, common-unit-of-account Nash equilibria exist without the mandate, so the mandate can operate as an equilibrium selection device. For $\rho \le 1-\varphi$, the symmetric mandate with $\hat{\mathbf{y}}_a = \hat{\mathbf{y}}_m$ selects an efficient outcome.*

(c) *For $\rho > \rho_c$, no common-unit-of-account Nash equilibrium exists without the mandate. Under the mandate with $\hat{\mathbf{y}}_a = \hat{\mathbf{y}}_m$, both banks face $\epsilon_{j,m} = 1/(1-\rho)$, and investment under the mandate falls strictly below n^* whenever $\epsilon_{j,m} > 1/\varphi$, i.e. $\rho > 1-\varphi$.*

(d) *Among mandates $\hat{\mathbf{y}}_a = \omega\hat{\mathbf{y}}_A + (1-\omega)\hat{\mathbf{y}}_B$, $\omega \in [0,1]$, the larger of the banks' settlement requirements for $\hat{\mathbf{y}}_a$, $\max\{\epsilon_{A,a}(\omega), \epsilon_{B,a}(\omega)\}$, is minimised at $\omega = 1/2$.*

Proof. Part (a): with $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k$, both constraints (9) and (10) carry the same unit of account. Setting $\chi_{j,k} > 0$ relaxes only (10), which (9) implies, while tightening the participation constraint, so $\chi^* = 0$. Part (b) follows from Proposition 6: common-unit-of-account equilibria exist for $\rho \le \rho_c$, and the symmetric common basket is efficient for $\rho \le 1-\varphi$ since $1/(1-\rho) \le 1/\varphi$. Part (c): Proposition 6 establishes that no common-unit-of-account equilibrium exists for $\rho > \rho_c$. Under the mandate with $\hat{\mathbf{y}}_a = \hat{\mathbf{y}}_m$, both banks face $\epsilon_{j,m} = \max_s\{\mathbf{p}_s'\hat{\mathbf{y}}_m/\mathbf{p}_s'\hat{\mathbf{y}}_j\} = 1/(1-\rho)$. When $\epsilon_{j,m} > 1/\varphi$ (i.e. $\rho > 1-\varphi$), the promise-keeping constraint binds and investment under the mandate falls below n^*. Part (d): by symmetry in the two-state model, $\max\{\epsilon_{A,a}(\omega), \epsilon_{B,a}(\omega)\}$ is minimised at $\omega = 1/2$. □

Proof of Proposition 9

Proof. Part (a): in the two-state model, $\epsilon_{j,m} = \max_s\{\mathbf{p}'_s\hat{\mathbf{y}}_m/\mathbf{p}'_s\hat{\mathbf{y}}_j\} = 1/(1-\rho)$. Since $(1+\rho)/(1-\rho) > 1/(1-\rho)$ for $\rho > 0$, the cash settlement requirement is strictly lower than the natural-unit direct requirement. Under the natural unit of account, fee-free cash settlement gives $B_j = n_j/H$ and requires $n_j^{\varphi} \geq eB_j$, where $e = \epsilon_{j,m} \geq 1$, which also ensures equation (9). Maximising $n_j^{\varphi} - n_j/H$ gives $W^m(e) = W^*$ for $e \leq 1/\varphi$ and $W^m(e) = (H/e)^{\frac{\varphi}{1-\varphi}}(1-1/e)$ otherwise. This is the direct-transfer problem with the fee fixed at zero, so $W^m(e) \leq W_j(e)$, with equality for $e \leq \bar{\epsilon}^{(0)}$ by Lemma C.1. Offering both channels cannot improve on their separately optimised payoffs: a positive direct fee induces cash use, while at zero fees equation (41) selects the smaller settlement requirement. The bank therefore earns $\max\{W_j(\epsilon_{j,k}), W^m(\epsilon_{j,m}), V_j^{(0)}\}$. For part (b), $\rho \leq \rho_c$ implies $\epsilon_{j,m} \leq \bar{\epsilon}^{(0)}$, so $W^m(\epsilon_{j,m}) = W_j(\epsilon_{j,m})$. Since $\epsilon_{j,m} \leq \epsilon_{j,k}$, Proposition 2 implies that cash weakly dominates direct transfer, strictly when $\rho > \rho_0$. Moreover, $W^m(\epsilon_{j,m}) \geq W_j(\bar{\epsilon}^{(0)}) > V_j^{(0)}$ by Proposition 3. The efficient-equilibrium existence boundary $1-\varphi$ is unchanged: efficient equilibria require $1/(1-\rho) \leq 1/\varphi$, and cash does not relax this condition. Under a common unit of account, $\hat{\mathbf{b}}_j = \hat{\mathbf{b}}_k$ implies $\epsilon_{j,k} = \epsilon_{j,j}$, so the interbank constraint is no tighter than the promise-keeping constraint; cash can relax the former but not the latter, leaving the binding constraint unchanged. Part (c): substituting $e = 1/(1-\rho)$ gives $W_m(\rho) = W^*$ for $\rho \leq 1-\varphi$ and $W_m(\rho) = \rho[H(1-\rho)]^{\frac{\varphi}{1-\varphi}}$ otherwise. This payoff is continuous, strictly decreasing from $W^* > V_j^{(0)}$ to zero on $[1-\varphi, 1)$, and hence crosses $V_j^{(0)}$ uniquely at $\rho_m^{(2)}$. Together with Proposition 3, this gives the combined threshold $\rho_c^{(2)} = \max\{\rho^{(2)}, \rho_m^{(2)}\}$, where autarky and cross-region trade are equally optimal. Part (d): when $\hat{\mathbf{b}}_k = \hat{\mathbf{y}}_m$, bank j's interbank parameter is $\epsilon_{j,k} = \max_s\{\mathbf{p}'_s\hat{\mathbf{y}}_m/\mathbf{p}'_s\hat{\mathbf{y}}_j\} = \epsilon_{j,m}$, so any cash contract can also be implemented by direct settlement with a zero fee and the same own-bank constraint; cash therefore offers no additional benefit. Bank k faces $\epsilon_{k,j} = \max_s\{\mathbf{p}'_s\hat{\mathbf{b}}_j/\mathbf{p}'_s\hat{\mathbf{y}}_k\}$, which may exceed $\epsilon_{k,m}$. Cash allows bank k to settle at $\min\{\epsilon_{k,j}, \epsilon_{k,m}\}$, strictly reducing its settlement requirement if and only if $\epsilon_{k,j} > \epsilon_{k,m}$. □

F. Proofs for Section 6

Proof of Proposition 10

Proof. Write $t = \varphi/(1-\varphi)$, $G = 1+\gamma$, $d = 1-\rho$ and $T = n_j^{\varphi}/B_j$. Equations (9) and (10) require $T \geq \epsilon_{j,j}$ and, if $I_F = 1$, $T \geq u_F\epsilon_{j,k}$, and participation gives $n_j^{\varphi} \leq (A/T)^t$, so positive profit is at most $(A/T)^t(1-c/T)$. An active channel with weight $\alpha_h \in \{\alpha, \beta\}$ adds $\alpha_h u_h$ to c and at most $\alpha_h G u_h$ to A, and an inactive one adds α_h to both, so $c \leq 1$

and $A \leq Gc - \gamma r$. Hence $A/c \leq G - \gamma r/c \leq H$, with equality only if both channels are active with zero fees and $\sigma_j = \sigma_k = 0$, since $\alpha, \beta, r > 0$. As $B_j \geq n_j/A$, profit is at most $\max_n\{n^\varphi - cn/A\} = (1-\varphi)(\varphi A/c)^t \leq W^*$. For part (i), at the midpoint profile $\epsilon_{j,j} = \epsilon_{j,k} = 1/d \leq 1/\varphi$, so each bank attains W^* by Proposition 2(i); as $\rho > 0$, $\sigma_j = 0$ only at the midpoint, so by strict concavity in n_j every other contract earns strictly less.

For part (ii), fix bank k at the midpoint, so $g_F = G$ and $\epsilon_{j,k} = 1/d > 1/\varphi$. If $T \geq 1/d$, then $T - c > c/t$, so $((Gc - \gamma r)/T)^t(1 - c/T)$, which bounds profit, rises in c; at $c = 1$ it equals $(H/T)^t(1 - 1/T)$, which falls in $T > 1/\varphi$ and equals $\rho(Hd)^t = W_m(\rho)$ at $T = 1/d$. The midpoint contract attains this payoff with both settlement constraints binding, $n_j = (Hd)^{1/(1-\varphi)}$ and $B_j = n_j/H$; fee-free cash in Appendix E is a distinct arrangement facing the same bound $T \geq 1/d$ with $A = H$ and $c = 1$. A deviation without cross-region trade earns at most $V^I(\rho)$.

Otherwise $T < 1/d$ and $I_F = 1$. Since $\epsilon_{j,j} \geq 1/d$ unless bank j's basket lies strictly on its own side of the midpoint, where $\epsilon_{j,j} = (1 - \sigma_j)/d$ and $g_L = G(1 - \sigma_j)$, and A rises in $1 - \sigma_j$ while c does not, set $1 - \sigma_j = dT \equiv 1/x$. Cross-region trade and equation (10) give $1/G \leq u_F \leq 1/x$, so $1 < x \leq G$, and inactive local trade is weakly dominated by trade at the fee with $g_L u_L = 1$, which leaves A unchanged and does not raise c. With $v = u_F x$, positive profit is at most $a^t(1 - b)$, where $a = d[rx + \alpha G u_L + \beta G v]$ and $b = d[rx + \alpha u_L x + \beta v]$, on the convex set given by $1 \leq x \leq G$, $x/G \leq u_L \leq 1$ and $x/G \leq v \leq 1$. Let $y = t\rho/(dH)$. On this set $(u_L - 1/G)(x-1) \geq 0$ and $(1-u_L)(G-x) \geq 0$, so replacing $u_L x$ in b by $u_L + (x-1)/G$ if $y \leq 1$, or by $Gu_L + x - G$ if $y > 1$, gives an affine $\bar{b} \leq b$, and $\bar{W} = a^t(1-\bar{b})$ bounds positive profit. Where $\bar{b} < 1$, which includes every contract with positive profit, $\log \bar{W}$ is concave; at $(x, u_L, v) = (1, 1, 1)$, $\bar{W} = W_m(\rho)$ and the derivatives of $\log \bar{W}$ in u_L, x and v, multiplied by ρ/d, are $\alpha(Gy - m_u)$, $ry - r - \alpha m_x$ and $\beta(Gy - 1)$, with $(m_u, m_x) = (1, 1/G)$ if $y \leq 1$ and $(G, 1)$ if $y > 1$. For $1 - \varphi < \rho \leq \hat{\rho}$, $1/H < y \leq (1-\beta)/r$, so $Gy > 1$, and either $y \leq 1$ and $ry \leq r$, or $Gy > G$ and $ry \leq r + \alpha$. As the set lies in $\{x \geq 1,\, u_L \leq 1,\, v \leq 1\}$, concavity gives $\bar{W} \leq W_m(\rho)$ wherever $\bar{b} < 1$. Hence, if $\rho \leq \hat{\rho}$ and $W_m(\rho) \geq V^I(\rho)$, the midpoint contract is a best response, and autarky is equally optimal when $W_m(\rho) = V^I(\rho)$.

Conversely, autarky is a feasible deviation, so $W_m(\rho) \geq V^I(\rho)$ is necessary. Along $u_L = v = 1$, which is feasible for x slightly above one with both settlement constraints binding, profit equals $a^t(1 - b)$, whose derivative in x at $x = 1$ has the sign of $ry - 1 + \beta$, positive when $\rho > \hat{\rho}$. Finally, since $g_L \leq G$ and an inactive local channel can be activated at a zero fee in the problem of Proposition 1, $V^I \leq V_j^{(0)}$, with equality for $\rho \leq 1 - \varphi$, where the midpoint has $\sigma_j = 0$ and $\epsilon_{j,j} = 1/d \leq 1/\varphi$. □

Lemma: Denomination Choice

Lemma F.1 (Denomination Choice)**.** *In the symmetric two-state model under the worst-state acceptance requirement, if local trade is active, a best response satisfies* $\sigma_j = 0$ *or* $\epsilon_{j,j} = n_j^{\varphi}/B_j$.

Proof. Suppose $\sigma_j > 0$ and $\epsilon_{j,j} < n_j^{\varphi}/B_j$. Moving $\hat{\mathbf{b}}_j$ slightly towards $\hat{\mathbf{y}}_m$ lowers $\sigma_j = \rho|1-2\omega|$, keeps equation (9) satisfied and leaves σ_k, $\epsilon_{j,k}$ and c unchanged. Since $\alpha > 0$ and local trade is active, A rises, so participation becomes slack and reducing B_j raises profit without violating equation (9) or (10). □

Reserves and Information-Insensitive Debt

Proposition F.1 (Reserves and Information-Insensitive Debt)**.** *Consider the two-state model with* $\rho > 1-\varphi$ *and a reserve requirement* τ *as in Proposition 8. If* $\tau \geq (\rho+\varphi-1)/\rho$*, the effective dispersion* $\rho(1-\tau)$ *satisfies* $\rho(1-\tau) \leq 1-\varphi$*, and the midpoint profile is a Nash equilibrium with* $\sigma_j = 0$ *that attains* W^*.

Proof. With $D = 1-\rho(1-\tau) \geq \varphi$, the midpoint has $\sigma_j = 0$, and its own-bank and cross-region settlement requirements relative to the effective resource basket both equal $1/D \leq 1/\varphi$, so the midpoint profile attains W^* by Proposition 2(i). Reserves leave expected resources and participation unchanged, so the bound W^* in the proof of Proposition 10 applies to every deviation. The argument does not substitute $\rho(1-\tau)$ for ρ in Proposition 10, which would require reparameterising baskets on the hull of the effective resource baskets. □